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MASTER'S FINAL WORK
DISSERTATION

COST-ADJUSTED BAYESIAN MODEL SELECTION
IN THE PRESENCE OF FACTORS

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Acknowledgments

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Abstract

Medical procedures often entail significant costs in terms of time, financial resources, specialized equipment, required expertise, or a combination of these. Traditional variable selection methods, focused solely on statistical fit (a “benefit-only analysis”), typically overlook the substantial variation in data-collection costs across predictors. As a result, selected models may include variables whose costs outweighs their predictive contribution, limiting their usefulness in resource-constrained clinical settings.

This work addresses the variable selection problem in logistic regression from a cost-benefit perspective when the pool of potential predictors comprises numerical variables and factors (categorical variables), some of which may benefit from reduced collection costs when used together (discounted predictors).

A Bayesian model selection framework is adopted, by defining a set of models where each element corresponds to a distinct subset of active variables. To guide the selection process, posterior model probabilities are computed using a prior structure that flexibly penalizes costly predictors, allowing practitioners to control the desired trade-off between predictive performance and data-collection burden according to the clinical context. This results in a cost-sensitive evaluation of competing models.

To properly account for multiplicity arising from the inclusion of factors, the findings of García-Donato and Paulo (2022b) are extended to Generalized Linear Models and, more specifically, to this cost-adjusted setting by using the g-priors proposed by Li and Clyde (2018) for model-specific parameters. Building upon the frameworks of Fouskakis, Ntzoufras, and Draper (2009b) and Porter, Franck, and Adams (2024), structured cost dependencies among predictors are incorporated, thus reflecting reality more closely. When exhaustive enumeration of the model space is computationally infeasible, a Gibbs sampler is employed to explore models with high posterior probability.

The practical impact of the proposed methodology is demonstrated through simulation studies and an application to a Cleveland Clinic Foundation heart disease dataset. The dataset includes several candidate “sickness indicators” with heterogeneous data-collection costs recorded for each patient.

Keywords: Bayes Factor; Categorical Variables; Cost-Benefit Analysis; Discounted Predictors; g-priors; Gibbs Sampler.

Resumo

Os procedimentos médicos envolvem frequentemente custos significativos em termos de tempo, dinheiro, equipamento ou qualquer combinação destes. Os métodos tradicionais de seleção de variáveis, focados exclusivamente no desempenho preditivo, ignoram geralmente a variação substancial nos custos de recolha das possíveis variáveis. Deste modo, os modelos selecionados podem incluir variáveis cujo custo supera a sua contribuição, reduzindo a sua utilidade em contextos clínicos com recursos limitados.

Este trabalho aborda o problema de seleção de variáveis numa perspetiva de custo-benefício no contexto de regressão logística, quando o leque de potenciais variáveis explicativas engloba variáveis numéricas e categóricas, algumas das quais podem beneficiar de uma redução nos custos de recolha quando utilizadas em conjunto (variáveis com desconto).

É adotada uma abordagem bayesiana para a seleção de modelos, definindo um espaço de modelos onde cada um corresponde a um subconjunto distinto de variáveis ativas. Para auxiliar o processo de seleção, são calculadas as probabilidades *a posteriori* dos modelos, utilizando uma estrutura *a priori* que penaliza flexivelmente as variáveis em função do seu custo, permitindo que o profissional de saúde tenha controlo sobre o natural custo de oportunidade entre desempenho preditivo e custo de recolha dos dados, de acordo com o contexto clínico. Desta forma, a comparação entre modelos torna-se sensível aos seus custos.

Com vista a assegurar uma representação adequada das variáveis categóricas perante o problema das comparações múltiplas, os resultados teóricos de García-Donato e Paulo (2022b) são estendidos à classe dos Modelos Lineares Generalizados e, mais especificamente, ao contexto apresentado, recorrendo às *g-priors* propostas por Li e Clyde (2018) para os parâmetros específicos dos modelos. Partindo das abordagens de Fouskakis, Ntzoufras, e Draper (2009b) e Porter, Frank, e Adams (2024), são tidas em consideração relações de dependência entre os custos das variáveis, refletindo de forma mais fiel a realidade. Quando a enumeração exaustiva do espaço de modelos é computacionalmente inexecutável, é utilizado um amostrador de Gibbs para explorar modelos com elevada probabilidade *a posteriori*.

O impacto da metodologia proposta é demonstrado através de estudos de simulação e da sua aplicação a um conjunto de dados reais da *Cleveland Clinic Foundation* sobre o diagnóstico de doenças cardíacas. Este conjunto de dados inclui vários candidatos a “indicadores de doença”, com custos de recolha heterogêneos registados para cada paciente.

Palavras-chave: Fator de Bayes; Variáveis Categóricas; Análise Custo-Benefício; Variáveis com Desconto; *g-priors*; Amostrador de Gibbs.

Glossary

BIC Bayesian Information Criterion. 1, 5, 28–30, 32, 33, 35

CHIC Confluent Hypergeometric Information Criterion. 23, 40, 43

EB Empirical Bayes. 11, 12, 21

EDM Exponential Dispersion Model. 3, 4, 6, 9, 10, 18, 21, 22, 24, 40

FB Fully Bayes. 11, 12, 21

GLM Generalized Linear Model. 2–4, 6, 20, 22, 23, 40, 44

HPM Highest Probability Model. 13–15, 28, 35, 44

iL Integrated Laplace. 2, 21–23, 40

IRWLS Iterative Re-Weighted Least Squares. 18

LIM Local Information Metric. 22, 23, 35, 40, 43

MCMC Markov Chain Monte Carlo. 2, 14–16, 27, 41

MLE Maximum Likelihood Estimate. 21–23, 35, 44

MPM Median Probability Model. 13, 28, 35–37, 44

PIP Posterior Inclusion Probability. vii, viii, 13, 15, 24–26, 35, 36

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1 Introduction

Statistical models provide a formal, albeit necessarily simplified, description of an observed phenomenon. In a parametric regression setting this description takes the form of a relationship between a response variable and a set of explanatory variables, and the analyst is typically faced with a pool of candidate predictors whose relevance is uncertain. Deciding which of them to retain is the variable selection problem. Whatever the inferential paradigm adopted, the criterion guiding this decision is almost invariably a measure of how well each subset of predictors explains the response. In this sense, the traditional formulation amounts to a *benefit-only* analysis: candidate predictors are judged exclusively by their statistical contribution.

Implicit in this formulation is the assumption that all candidate predictors are equally easy to obtain. In many applications they are not. Collecting a variable may require money, time, specialized equipment, expert interpretation, or a combination of these, and such requirements often differ substantially across the candidate pool. Medical diagnosis is a paradigmatic example: recording a patient’s resting blood pressure is essentially costless once the patient is in the consulting room, whereas an exercise electrocardiogram study demands dedicated equipment, specialized staff and considerable time. A model achieving a marginally better fit at a substantially higher per-patient cost may therefore be of little practical use in a resource-constrained clinical setting. Besides, the “gold standard” procedure for diagnosing a given condition may simply be unaffordable as a routine tool, or too invasive to justify its use outside cases where it is strictly necessary, prompting the search for alternatives that are cheaper, less invasive, or both. In such circumstances the goal is not the best-fitting model, but a subset of predictors delivering comparable diagnostic performance at a materially lower data-collection burden.

Bayesian model selection offers a natural framework for this kind of analysis. Competing models are compared through their posterior probabilities, which combine the evidence the data provide for each model with the prior weight assigned to it. Cost information can thus be incorporated without altering the sampling model, simply by letting the prior over the model space penalize expensive predictors. This is the route taken by Fouskakis, Ntzoufras, and Draper (2009b), whose prior induces posterior model odds that generalize the Bayesian Information Criterion (BIC), and subsequently refined by Porter, Franck, and Adams (2024), who introduced a single user-controlled tuning parameter governing the strength of the cost penalty. This flexibility matters because practitioners rarely face the same scenario: resource constraints vary from one clinical setting to another, and so does the level of predictive performance they are willing to accept in exchange for a lower data-collection burden.

Two limitations of the existing cost-penalized literature motivate this dissertation. The first concerns the nature of the candidate predictors: available methods assume that all predictors are numerical or, when factors are present, treat each one as an indivisible block, entering the model through a single inclusion decision, which may be restrictive (García-Donato & Paulo, 2022b). Here, a factor is regarded as relevant whenever at least one of the coefficients associated with its levels

is non-zero. Following García-Donato and Paulo (2022b), the class of competing models must be built so that every combination of active levels is entertained, and the prior over that class must (i) keep the prior inclusion probability of the factor as a whole independent of its number of levels, and (ii) control for multiplicity both across predictors and within the group of dummy variables representing a factor. The second limitation concerns the cost structure itself. Data-collection costs have so far been treated as predictor-specific and mutually independent, whereas in practice a single procedure frequently yields several variables at once, termed grouped predictors: once a blood sample has been drawn, the additional cost of obtaining further biomarkers from it is negligible. Likewise, other groups of predictors may share equipment, personnel, or preparation steps that make them cheaper to collect jointly than independently; these are referred to here as discounted predictors. Ignoring these dependencies systematically overstates the true data-collection burden of a model, thereby reducing the practical relevance of the resulting variable selection procedure.

This work addresses both limitations. On the one hand, the model space construction and the two-stage model space priors of García-Donato and Paulo (2022b) are implemented for Generalized Linear Models (GLMs), using the g-priors of Li and Clyde (2018) for the model-specific parameters and Integrated Laplace (iL) approximations for the resulting marginal likelihoods. The invariance of the approximate Bayes factors to the choice of reference levels, which justifies this construction, is established, and the resulting closed-form expressions make it possible to explore the model space with a simple Markov Chain Monte Carlo (MCMC) algorithm, the Gibbs sampler. On the other hand, the cost-adjusted model prior is extended so that a factor is penalized as a whole rather than level by level, keeping prior inclusion probabilities free of the number of levels, and so that structured cost dependencies, in the form of discounted and grouped predictors, can be accommodated.

The remainder of this dissertation is organized as follows. Chapter 2 reviews the theoretical foundations of the variable selection problem from the perspective of Bayesian model selection. Chapter 3 extends the methodology of García-Donato and Paulo (2022b) for variable selection in the presence of categorical predictors from linear regression models to non-normal GLMs, under the g-priors for model-specific parameters proposed by Li and Clyde (2018). The chapter also presents two simulation studies examining the recommended choice of model space prior. Chapter 4 turns to the non-traditional setting of cost-penalized variable selection, builds on the framework of Fouskakis, Ntzoufras, and Draper (2009b) and Porter, Franck, and Adams (2024), and further extends it to accommodate categorical, grouped, and discounted predictors in cost-benefit scenarios. The proposed methodology is then applied to a real dataset to select cost-effective predictors for the diagnosis of heart disease. Chapter 5 summarizes the key findings and their implications, discusses limitations and challenges encountered, and outlines directions for future research.

2 Literature Review

Parametric statistical models provide a formal probabilistic framework for interpreting observed phenomena in a necessarily simplified way, with the goal of enabling statistical inference about the underlying probability distribution. From observed data, they allow us to describe past behavior and make predictions about future outcomes of similar phenomena (Robert, 2007).

2.1 Exponential Dispersion Model Family

A continuous (discrete) random variable $y \sim \text{EDM}(\mu, a(\phi))$, with mean μ and $a(\phi) > 0$, belongs to the Exponential Dispersion Model (EDM) family of distributions if its probability density (mass) function, p.d.f. (p.m.f.), is of the form (McCullagh & Nelder, 1989):

$$f(y \mid \theta, \phi) = \exp \left[\frac{y\theta - b(\theta)}{a(\phi)} + c(y, \phi) \right]. \quad (1)$$

(1) is the canonical form of an EDM, as it is written in terms of the *canonical parameter* θ . Other parameterizations are also possible, including the *dispersion model form*, written in terms of μ (see Ch. 5 of P. K. Dunn and Smyth, 2018).

Here, $c(y, \phi_0)$ is a *normalizing term*, ensuring that the p.d.f. integrates (or the p.m.f. sums) to 1. The function $b(\theta)$, known as the *cumulant function*, plays a fundamental role in defining the properties of the distribution. For EDMs, it follows that: $\mathbb{E}(y) = \mu = b'(\theta)$ and $\mathbb{V}(y) = a(\phi)b''(\theta) > 0$, establishing a one-to-one relationship between the mean μ and θ . The second derivative, $b''(\theta) = V(\mu)$, known as the *variance function*, expresses the mean-variance relationship of the data. It is particularly important because it uniquely identifies the distribution within the EDM family, since $V(\mu)$ determines $b(\theta)$ up to an additive constant, and this in turn specifies the *cumulant generating function*, $K(t) = \log M(t) = \log \mathbb{E}[e^{ty}]$ (see Ch. 5 of P. K. Dunn and Smyth, 2018 and Ch. 2 of Casella and Berger, 2002). Notably, the EDM family includes many common continuous (e.g., normal and gamma) and discrete (e.g., poisson and binomial) distributions.

2.2 Generalized Linear Models

GLMs are a class of statistical models used for regression tasks, i.e., modeling the relationship between a response variable, $\mathbf{Y}$, continuous or discrete; and a set of k potential explanatory variables $\mathbf{x}_1, \dots, \mathbf{x}_k$. As with all parametric statistical models, they are defined by two components, described below:

- **Random Component:** The observations in $\mathbf{Y} = (y_1, \dots, y_n)^\top$ follow probability distributions from the EDM family, where $y_i \sim \text{EDM}(\mu_i, a(\phi))$, for $i = 1, \dots, n$. The choice of distribution depends on the nature of the response data or knowledge of how the variance relates with the mean. The *dispersion parameter*, ϕ , from the function $a(\phi) = \phi/w_i$ in (1), is assumed to be constant over the independent observations, while w_i is a known non-negative *prior weight* that may vary with i .

- **Systematic Component:** The linear predictor, $\boldsymbol{\eta} = (\eta_1, \dots, \eta_n)^\top$, is defined as a linear combination of the independent variables included in the model:

$$\boldsymbol{\eta} = \mathbf{1}_n \alpha + \mathbf{X} \boldsymbol{\beta} = \mathbf{1}_n \alpha + \mathbf{x}_1 \beta_1 + \mathbf{x}_2 \beta_2 + \dots + \mathbf{x}_k \beta_k, \quad (2)$$

where α is the intercept; $\mathbf{X} = [\mathbf{x}_1 \mid \dots \mid \mathbf{x}_k]$ the design matrix; $\mathbf{x}_j = (x_{1j}, \dots, x_{nj})^\top$ the n -dimensional vector of observations for the j -th explanatory variable, for $j = 1, \dots, k$; and α_j is the corresponding regression coefficient, so that $\boldsymbol{\beta} = (\beta_1, \dots, \beta_k)$. The response means, $\boldsymbol{\mu}$, are related to $\boldsymbol{\eta}$ through a monotonic (invertible) and differentiable link function,

$$\eta_i = h(\mu_i). \quad (3)$$

Monotonicity guarantees that each value of η_i is uniquely mapped to a corresponding μ_i , and consequently to θ_i . Hence, $y_1, \dots, y_n$ are not identically distributed, as their corresponding canonical parameters $\theta_1, \dots, \theta_n$ are linked with the predictors via $\theta_i = \theta(\eta_i)$. Under this parameterization, the canonical link is a special link function, such that $\theta_i = \eta_i$.

Thus, GLMs maintain linearity in the parameters while allowing flexibility in the response distribution and its relationship to explanatory variables, generalizing the well-known (normal) linear regression model, where $\eta = g(\mu) = \mu$ (P. K. Dunn & Smyth, 2018). Since the normal distribution is a member of the EDM family, this model can be expressed as $y_i \sim \mathcal{N}(\mu_i, \phi/w_i)$ or, equivalently, as $y_i - \mu_i \sim \mathcal{N}(0, \sigma^2/w_i)$, where ϕ is denoted by σ^2 . This follows because subtracting a constant to a normal random variable only affects its location parameter (Casella & Berger, 2002). Defining $\varepsilon_i = y_i - \mu_i$ recovers the familiar linear regression formulation with independent and identically distributed errors for the homoskedastic case $w_i = 1$ for all i (Wooldridge, 2013).

2.3 The Variable Selection Problem

Consider the model with systematic component given in (2). Referred to as the full model, $\mathcal{M}_F$, it includes all potential explanatory variables, which are assumed to be numerical throughout this chapter. The variable selection problem consists of determining which of these regressors are relevant for explaining the response variable, that is, identifying the “best” predictor subset for $\mathbf{Y}$. There are two inferential approaches to address this problem (García-Donato & Paulo, 2022b). The first consists of (point) estimation methods, in which only the full model $\mathcal{M}_F$ is taken into account. The second approach involves model selection methods, where inference is conducted by considering the set of all possible models, $\mathcal{M}$, defined by the different subsets of the candidate variables ($|\mathcal{M}| = 2^k$, if considering only main effects). Model selection methods fall short in terms of computational feasibility (especially if k is large), since estimation techniques require fitting only a single model. In contrast, model selection results enable the production of model-averaged

estimates, offering a curious perspective on the problem (Fernández, Ley, & Steel, 2001; Fletcher, 2018).

The two dominant paradigms of statistical inference, Frequentist and Bayesian, provide different solutions to addressing the problem of variable selection, within each of these approaches. The fundamental distinctions between these schools of thought stem from their treatment of model parameters, their conceptualization of probability, and their methods for quantifying uncertainty. Frequentist inference treats the regression coefficients as fixed but unknown quantities. Within this framework, regression parameters are estimated solely from the observed data using an appropriate estimator (Friedman, Hastie, & Tibshirani, 2010; Wooldridge, 2013). The resulting point estimates are then used for prediction, without explicitly accounting for the uncertainty associated with estimating the unknown parameters. Uncertainty is instead attributed to the data-sampling process and is assessed through the sampling distribution of the estimators, when available, or through resampling techniques such as the bootstrap. See, for instance, Desboulets (2018) for a detailed overview of Frequentist methods. On the other hand, Bayesian inference treats the regression coefficients as random variables. A prior distribution is specified to represent beliefs about the parameters before observing the data. After the data are observed, the prior is updated through Bayes' theorem to obtain the posterior distribution for the regression coefficients. Inference is then based on the (full probability) posterior distribution conditional on the observed data. This provides a fully probabilistic interpretation of parameter uncertainty. Further details on Bayesian estimation methods are provided in Castillo, Schmidt-Hieber, and van der Vaart (2015), while (bayesian) model selection methods are discussed in the remainder of this work.

In estimation methods, the assumed (full) model most likely contains many variables with uncertain effects on the response, making dimensionality reduction desirable, but challenging. Regression parameters can be encouraged to be zero either through penalty-based methods, in the Frequentist framework or via prior distributions that place substantial mass near zero on the regression parameters in the Bayesian framework. In contrast, model selection methods naturally promote sparse representations. In model selection techniques, models of different dimensions are considered. In particular, models simpler than $\mathcal{M}_F$ are taken into account. When comparing nested or non-nested models within the Frequentist framework, criteria such as BIC are commonly used (P. K. Dunn & Smyth, 2018, Ch. 2). These criteria balance model fit and parsimony by applying penalties for model complexity, thereby favoring simpler models, *ceteris paribus*. In the Bayesian framework, a similar preference for parsimonious models arises naturally through the *marginal likelihood*. This quantity plays a central role in Bayesian model comparison and will be discussed in detail later.

The implementation of model selection methods also raises some theoretical issues. In particular, these methods must account appropriately for the effects of multiple comparisons. Because each candidate model has a nonzero probability of being incorrectly identified as significant, the overall probability of committing at least one Type I error increases as the number of models of similar complexity grows.

In the Bayesian paradigm, controlling for multiplicity has been a topic of interest since Jeffreys (1961), and we discuss this issue in greater detail in the next section. In the Frequentist paradigm, classical approaches include the procedures proposed by O. J. Dunn (1961) and Benjamini and Hochberg (1995). By contrast, estimation methods are not subject to the multiple-testing problem, as they fit a single model by construction.

2.4 Bayesian Model Selection

As outlined in the previous section, the Bayesian model selection approach to the variable selection problem provides a natural and general probabilistic framework that accounts for parameter uncertainty and, in addition, uncertainty surrounding the probabilistic model. The goal is to identify the model that, from an initial set of candidate models, $\mathcal{M}$, best explains the observed data, $\mathbf{Y}$, according to a specified criterion. All models in $\mathcal{M}$, the model space, are assumed to share the same sampling distribution family, which in this work is taken to be a member of the EDM family of distributions. Therefore, competing models differ only in the subset of variables included as explanatory covariates for $\boldsymbol{\eta}$ in (2). This framework also accommodates situations in which certain variables are included in all models (e.g., due to model constraints), thereby reducing the effective size of the model space.

It is common to represent model uncertainty in $\mathcal{M}$ as uncertainty in $\boldsymbol{\gamma} = (\gamma_1, \dots, \gamma_k)^\top$, a binary vector in $\{0, 1\}^k$ encoding the predictors included in a given model (e.g., George and McCulloch, 1993; M. A. Clyde and George, 2004). In this formulation, γ_j serves as an indicator variable taking the value 1 if predictor $\mathbf{x}_j$ is included in the model and 0 if it is not (equivalently, $\gamma_j = 1 \iff \beta_j \neq 0$), for $j = 1, \dots, k$. This means that under model $\mathcal{M}_{\boldsymbol{\gamma}}$, observations in $\mathbf{Y}$ are generated from

$$(y_i \mid \mu_i(\boldsymbol{\gamma}), \phi) \stackrel{\text{indep}}{\sim} \text{EDM} \left(\mu_i(\boldsymbol{\gamma}), \frac{\phi}{w_i} \right), \quad i = 1, \dots, n; \quad (4)$$

and the systematic component of the GLM is

$$\boldsymbol{\eta}_{\boldsymbol{\gamma}} = \mathbf{1}_n \alpha + \mathbf{X} \text{diag}(\boldsymbol{\gamma}) \boldsymbol{\beta} = \mathbf{1}_n \alpha + \mathbf{X}_{\boldsymbol{\gamma}} \boldsymbol{\beta}_{\boldsymbol{\gamma}}, \quad (5)$$

where each of the n entries $\eta_i(\boldsymbol{\gamma})$, of the linear predictor $\boldsymbol{\eta}_{\boldsymbol{\gamma}}$ is related with the corresponding response mean $\mu_i(\boldsymbol{\gamma})$ via (3). $\mathbf{X}_{\boldsymbol{\gamma}}$ is the model-specific design matrix with $k_{\boldsymbol{\gamma}} = \sum_{j=1}^k \gamma_j \leq k$ columns. These columns correspond to those of the full design matrix $\mathbf{X}$ associated with the nonzero elements of $\boldsymbol{\gamma}$. In other words, a variable $\mathbf{x}_j$ is in model $\mathcal{M}_{\boldsymbol{\gamma}}$ if and only if its corresponding column in $\mathbf{X}$ has not been deleted. The vector $\boldsymbol{\beta}_{\boldsymbol{\gamma}}$, of length $k_{\boldsymbol{\gamma}}$, contains the regression coefficients attached to the selected covariates in model $\mathcal{M}_{\boldsymbol{\gamma}}$. Note that the intercept α is included in every model in $\mathcal{M}$. Thus, let $\mathcal{M}_{\emptyset}$ denote the null model, with only the intercept α ($k_{\mathcal{M}_{\emptyset}} = 0$).

To account for uncertainty regarding the appropriate model, the natural solution in the Bayesian framework is to assign a prior distribution over the model

space. Models outside this space are implicitly assigned zero prior probability and are therefore excluded from the subsequent inference. We denote the prior probability of model $\mathcal{M}_\gamma$ by either $f(\mathcal{M}_\gamma)$ or $f(\gamma)$. Conditional on each candidate model, uncertainty about the unknown parameters is also represented through a prior distribution, $f(\boldsymbol{\psi}_\gamma | \mathcal{M}_\gamma)$, with $\boldsymbol{\psi}_\gamma = (\alpha, \boldsymbol{\beta}_\gamma, \phi) \in \boldsymbol{\Psi}_\gamma$. The model space prior, together with the joint prior distribution on the unknown parameters and the sampling model in (4), defines a hierarchical mixture framework encompassing the collection of candidate models. We also assume the link function $h(\cdot)$ is fixed and known. By contrast, Ntzoufras, Dellaportas, and Forster (2003) treated the link function as an unknown parameter, while also accounting for the uncertainty associated with variable selection.

Inference is not based solely on these prior opinions about the unknown quantities of interest. Instead, they are updated in light of the observed data, $\mathbf{Y}$, which are treated as fixed. This updating is naturally achieved through Bayes' theorem (Casella & Berger, 2002), yielding the posterior distribution. Consequently, the posterior probability of a single model $\mathcal{M}_\gamma$, conditional on the observed data, is given by

$$f(\mathcal{M}_\gamma | \mathbf{Y}) = \frac{f(\mathbf{Y} | \mathcal{M}_\gamma)f(\mathcal{M}_\gamma)}{\sum_{\mathcal{M}_\ell \in \mathcal{M}} f(\mathbf{Y} | \mathcal{M}_\ell)f(\mathcal{M}_\ell)} = \frac{f(\mathbf{Y} | \mathcal{M}_\gamma) f(\mathcal{M}_\gamma)}{C^{-1}}, \quad (6)$$

where C is the constant of proportionality. The quantity $f(\mathbf{Y} | \mathcal{M}_\ell)$ in (6) is referred to in the literature by several names (García-Donato & Paulo, 2022b; Li & Clyde, 2018; Porter, Franck, & Adams, 2024), including marginal likelihood (or integrated likelihood) of the data under model $\mathcal{M}_\ell$ and the prior predictive density of the data under the same model. Being one of the fundamental quantities used for model comparison, it can be formally defined as,

$$f(\mathbf{Y} | \mathcal{M}_\gamma) = \int_{\boldsymbol{\Psi}_\gamma} f(\mathbf{Y} | \boldsymbol{\psi}_\gamma, \mathcal{M}_\gamma) f(\boldsymbol{\psi}_\gamma | \mathcal{M}_\gamma) d\boldsymbol{\psi}_\gamma, \quad (7)$$

obtained by integrating the likelihood of the observed data $\mathbf{Y}$, with respect to the prior distributions of the unknowns. It should be noted that this constitutes a slight abuse of terminology. Throughout this work, the joint p.d.f (p.m.f) of the sampling model in (4), $f(\mathbf{Y} | \boldsymbol{\psi}_\gamma, \mathcal{M}_\gamma)$, is referred to as the *likelihood* under model $\mathcal{M}_\gamma$, although, strictly speaking, it is only proportional to the likelihood function $\mathcal{L}(\boldsymbol{\psi}_\gamma | \mathbf{Y}, \mathcal{M}_\gamma)$ (Casella & Berger, 2002). This distinction arises from the fact that the likelihood has no intrinsic normalization and is therefore defined only up to a positive multiplicative constant, which cancels when computing (6), so that this abuse of terminology has no practical consequences.

Equally important is the null-based Bayes Factor (Kass & Raftery, 1995) for

comparing $\mathcal{M}_\gamma$ to the common baseline model $\mathcal{M}_\emptyset$

$$\text{BF}_{\mathcal{M}_\gamma:\mathcal{M}_\emptyset} = \frac{f(\mathbf{Y} \mid \mathcal{M}_\gamma)}{f(\mathbf{Y} \mid \mathcal{M}_\emptyset)}. \quad (8)$$

In the context of Bayesian parametric model comparison, each competing hypothesis is represented by a statistical model. Therefore, this Bayes factor quantifies the relative support that the observed data provide for model $\mathcal{M}_\ell$ over the null model $\mathcal{M}_\emptyset$; larger values indicate stronger evidence in favor of $\mathcal{M}_\ell$. In the Bayesian framework, Jefferys and Berger (1992) highlighted the Bayesian Ockham’s razor effect, providing quantitative support for the principle that, among models consistent with the data, simpler models are generally preferred. This effect arises because the marginal likelihood (7) contains an inherent penalty for model complexity. Specifically, the likelihood is integrated over the parameter space, which is higher dimensional for more complex models. Intuitively, such models tend to assign less prior probability on the parameters (variables) that “actually” explain the data, since the joint prior on regression parameters integrates to 1 over the entire parameter space. As a result, the predictive distribution for the data (7) tends to be more diffuse for more complex models. Consequently, the Bayes factor in (8) naturally induces sparsity, *ceteris paribus*.

Implementing Bayesian methods presents important challenges, which we discuss next. These include the specification of the prior distributions $f(\mathcal{M}_\gamma)$ and $f(\boldsymbol{\psi}_\gamma \mid \mathcal{M}_\gamma)$; the evaluation of the marginal likelihood (7), owing to its dependence on the latter prior specification and the form of the likelihood (4); and computing posterior model probabilities (6), hereafter denoted interchangeably with $f(\gamma \mid \mathbf{Y})$ when the model space is large (i.e., for high values of k).

2.4.1 Priors for model parameters

As with any other Bayesian approach, prior distributions for model parameters encode prior knowledge or assumptions before the data are observed. In practice, however, such priors often do not reflect genuine prior information. While expert knowledge can be used to construct informative priors, it may be unavailable or difficult to elicit. This challenge is particularly pronounced when many parameters must be specified, since experts are unlikely to reach a consensus. Consequently, subjective prior elicitation also becomes impractical (Hoff, 2009, Ch. 9). Resorting to frequentist methods would forgo the advantages offered by the Bayesian framework (see section 1.2 of Berger and Pericchi, 2001). Instead, one can rely on *default* prior distributions. These priors are motivated by formal methods (Kass & Wasserman, 1996) and are chosen to be as minimally informative as possible. Consequently, the resulting posterior information is driven primarily by the observed data, while reflecting only weak, neutral prior assumptions. Although such priors are not entirely free of assumptions, they are often regarded as providing a “more objective” basis for inference.

The literature on objective prior specification for $f(\boldsymbol{\psi}_\gamma \mid \mathcal{M}_\gamma)$ is extensive. In

this work, we adopt conventional priors, following the terminology introduced by Berger and Pericchi (2001). Conventional priors have been widely adopted due to their computational tractability, particularly in the context of linear regression, the canonical framework for model uncertainty and variable selection (Liang et al., 2008). In this setting, Bayarri et al. (2012) showed that this class of priors satisfies several desiderata, establishing it as a compelling choice for objective Bayesian model selection. Under the more general scope of the EDM family of distributions, the class of priors proposed by Li and Clyde (2018), which underlies the developments presented in this dissertation, also satisfies several of these desiderata in a broad range of cases.

Following the principles introduced by Jeffreys (1961), the conventional approach assigns improper, non-informative priors to parameters common to the models, provided certain conditions are met (as discussed in the next chapter), and proper default priors to the remaining parameters. This leads to coherent prior specifications for a pair of nested hypotheses (Zellner & Siow, 1980), where the null hypothesis, H_0 , constrains values of β_γ . This justifies computing null-based Bayes factors in (8), as the null model $\mathcal{M}_\emptyset$ is nested within every other model in $\mathcal{M}$. Consequently, each pair of nested models shares the same common parameters: the intercept α and the dispersion parameter ϕ . Then, for any non-null model $\mathcal{M}_\gamma \neq \mathcal{M}_\emptyset$ in $\mathcal{M}$, $f(\psi_\gamma | \mathcal{M}_\gamma)$ can be factorized, without loss of generality, as $f(\beta_\gamma | \phi, \mathcal{M}_\gamma) f(\alpha, \phi)$; while for the null model $\mathcal{M}_\emptyset$ only the prior $f(\alpha, \phi)$ is required. Nonetheless, any pair of models in $\mathcal{M}$, whether nested or not, can still be compared through the ratio of their null-based Bayes factors. Specifically, for any two models $\mathcal{M}_r$ and $\mathcal{M}_s$ in $\mathcal{M}$, $\text{BF}_{\mathcal{M}_r:\mathcal{M}_s} = \text{BF}_{\mathcal{M}_r:\mathcal{M}_\emptyset} / \text{BF}_{\mathcal{M}_s:\mathcal{M}_\emptyset}$.

Improper priors are not probability distributions in the usual sense, as their densities do not integrate to one (Wasserman, 2004, Ch. 11). Instead, they are defined only up to a multiplicative constant. Nevertheless, they may still be used in Bayesian inference, because the associated multiplicative constant is identical under every model and therefore cancels when computing the null Bayes factor in (8). The same argument does not apply to priors on model-specific parameters β_γ , for which the use of improper priors is generally inadmissible. The arbitrary multiplicative constants associated with such priors differ across models and, consequently, would lead to indeterminate Bayes factors (Kass & Raftery, 1995). This requirement was formalized by Bayarri et al. (2012) as the *basic criterion*, which states that the conditional priors $f(\beta_\gamma | \mathcal{M}_\gamma)$ must be proper. The arguments of Berger and Pericchi (2001) further exclude the use of overly vague proper priors for β_γ .

Among the conventional class of prior distributions, Zellner’s (1986) g-prior,

$$\beta_\gamma | \phi, \mathcal{M}_\gamma \sim \mathcal{N}_{k_\gamma}(\mathbf{0}, g\phi(\mathbf{X}_\gamma^\top \mathbf{X}_\gamma)^{-1}), \quad (9)$$

is perhaps the most popular. Coefficients not included in $\mathcal{M}_\gamma$ are fixed at zero through a point mass, so that priors of this type are also referred to as point-mass spike-and-slab priors, in contrast to the continuous spike-and-slab priors of George and McCulloch (1993). The null-based specification in (9) centers the symmetric prior at the origin, and is only one of several variants of the g-prior proposed in the

literature for (homoskedastic) Gaussian regression models. Therefore, elicitation collapses to the single hyperparameter $g > 0$, which governs the prior dispersion. The appeal of (9) lies in this structure, a multiple of the variance of the Maximum Likelihood Estimator of β_γ , or equivalently, a multiple of the inverse of the expected Fisher information matrix (P. K. Dunn & Smyth, 2018). This structure ensures model selection is invariant to changes in the scales of the regressors (Hoff, 2009). Additionally, it allows the integral in (7) to be rapidly evaluated in closed form, at the cost of a single least-squares fit (Liang et al., 2008). One of the reasons for this is that the normal prior in (9) is conjugate (Wasserman, 2004) to the normal likelihood.

Outside the normal case, however, the whole construction loses its analytical footing. For the remaining EDMs, normal priors on β_γ are no longer conjugate to the likelihood $f(\mathbf{Y} \mid \psi_\gamma, \mathcal{M}_\gamma)$, and the integral in (7) admits no closed-form solution. Although algorithms exist that circumvent this intractability without requiring a closed-form expression (see, e.g., Ntzoufras, Dellaportas, and Forster, 2003; and George and McCulloch, 1993), these are often time-consuming, undesirable for efficient computation of posterior model probabilities, as they involve sampling from the posterior distribution of the model parameters. The alternative, pursued in this work, is to use normal g-priors similar to (9) and approximate the likelihood analytically by Laplace’s method (Kass & Raftery, 1995), described in the next chapter.

2.4.2 Model Space Priors

The construction of model space priors has long been an active area of research. One line of work has focused on developing priors with desirable properties within an objective Bayesian framework across a wide range of traditional model selection settings (see, e.g., Denison, Mallick, and Smith, 1998; George and Foster, 2000; Yang, Wainwright, and Jordan, 2016). In the following paragraphs particular emphasis is placed on the multiple-testing problem. Scott and Berger (2010) argued that multiplicity adjustment arises from an appropriate choice of prior probabilities over models, rather than from the marginal likelihoods. Thus, unlike the Bayesian Ockham’s razor effect, multiplicity adjustment is not an inherent feature of Bayesian model selection, but rather a consequence of the chosen model space prior.

In the simplest traditional setting, a natural default prior on the model space is

$$f(\gamma \mid \mathbf{w}) = \prod_{j=1}^k f(\gamma_j \mid w_j) = \prod_{j=1}^k w_j^{\gamma_j} (1 - w_j)^{1-\gamma_j}, \quad (10)$$

where $\mathbf{w} = (w_1, \dots, w_k)$ and w_j denotes the prior probability of the j th variable being included in the model. Equivalently, $\gamma_j \mid w_j \sim \text{Ber}(w_j)$, $j = 1, \dots, k$. The factorization is justified because, in this setting, we assume there are no structured dependencies among the candidate variables (e.g., through interaction or higher-order terms), and no covariate redundancy or collinearity. For extensions to those settings, see H. Chipman, George, and McCulloch (2001), H. A. Chipman (1996),

and Taylor-Rodríguez, Womack, and Bliznyuk (2016). Furthermore, categorical predictors are not considered in the present framework; they receive more detailed treatment in the next chapter, where they constitute a primary focus.

Although each w_j could, in principle, be specified using prior knowledge (e.g., from expert opinion), such information is often unavailable or costly to elicit. The usual approach is to assume that variable inclusions are exchangeable Bernoulli trials with common inclusion probability w : $f(\boldsymbol{\gamma} \mid w) = w^{k_\gamma}(1-w)^{k-k_\gamma}$. Under this formulation, the size of a model, k_γ , follows a $\text{Bin}(k, w)$ distribution, since there are $\binom{k}{k_\gamma}$ models with k_γ regressors in $\mathcal{M}$. This reduces prior specification to the prior expected proportion of variables included in the model, w . Consequently, prior odds will favor larger models if $\mathbb{E}[k_\gamma] = kw > k/2$ and the opposite if $kw < k/2$. For the choice of $w = 0.5$ all models have equal prior probability $f(\boldsymbol{\gamma} \mid w = 0.5) = |\mathcal{M}|^{-1} = 2^{-k}$. This constant prior on the model space was initially regarded as the natural “noninformative” choice.

Note that this prior ignores the uneven number of models across dimensions: there are far more models with k_γ around $k/2$ than with k_γ close to 0 or k . Consequently, it induces a symmetric bell-shaped prior distribution on the model size, k_γ , around its prior mean, so that the prior odds favor models of intermediate dimension. This preference for models of moderate size has important implications as the number of candidate predictors increases. As spurious variables are added to the candidate set of variables, incorrect models are introduced and the number of model comparisons increases. Consequently, the probability of incorrectly identifying a model as significant also rises, since the prior probability assigned to the single true model decreases while more prior mass is distributed to the rest of models. Under the “pseudo-objective” constant prior, however, the prior distribution of the inclusion proportion becomes increasingly concentrated around $1/2$ as k grows, even though the additional variables are purely noise (Bogdan, Ghosh, & Tokdar, 2008). This behavior is consistent with the approximately linear increase in false positives (i.e., noise variables incorrectly identified as significant) observed in the experiments of Scott and Berger (2010). Similar conclusions hold for any other fixed choice of w that does not depend on k .

Two different approaches can be used to achieve automatic multiplicity adjustment, provided they are specified appropriately. In the simulation experiments of Scott and Berger (2010), this translated into keeping the number of false positives small despite the increase in the number of spurious variables. The first approach adopts an Empirical Bayes (EB) strategy by treating w as an unknown parameter to be empirically estimated from the data using a consistent estimator, most commonly obtained by (Type-II) maximum likelihood (Cui & George, 2008), thereby introducing the dependence on k . This estimated $\hat{w}$ is then used to determine the posterior distribution over the model space: $f(\boldsymbol{\gamma} \mid \mathbf{Y}) \propto \hat{w}^{k_\gamma} \cdot (1 - \hat{w})^{k-k_\gamma} \cdot f(\mathbf{Y} \mid \boldsymbol{\gamma})$. The multiplicity control arises because $\hat{w} \rightarrow 0$ as k grows while $k_{\mathcal{M}_T}$ remains fixed. As the underlying proportion of candidate variables included in the true model decreases, the prior bias against inclusion becomes progressively stronger, making it increasingly difficult for any variable to be selected.

The alternative approach, Fully Bayes (FB), treats w as a hyperparameter and

specifies a hierarchical model over the model space by placing a prior on w , $\pi(w)$, usually a Beta(a,b) distribution (Ley & Steel, 2009). This *hyperprior* induces a less informative prior on the model size and allows the prior to adapt more naturally to the observed data than the constant prior. It also down weights, *a priori*, models with k_γ close to $k/2$. Then, the prior on the model space can be written as $f(\gamma) = \int_0^1 f(\gamma \mid w) \pi(w) dw$. According to Scott and Berger (2010), the FB solution to multiplicity adjustment is obtained by choosing $a = b = 1$, which is equivalent to placing a Unif(0,1) prior on the *hyperparameter* w . We refer to the resulting hierarchical prior as the *SB prior*, which induces

$$f(\gamma) = (k+1)^{-1} \binom{k}{k_\gamma}^{-1}. \quad (11)$$

This prior assigns the same total prior probability to each model dimension and distributes it uniformly among all models of that dimension, so that the prior probability of a model is inversely proportional to the number of models of the corresponding dimension. Accordingly, the prior probability assigned to an individual model, viewed as a function of its size, follows a symmetric U-shaped curve, with peaks at model sizes 0 and k . This reveals the multiplicity adjustment inherent in the prior probabilities, which may either penalize or favor a model depending on its position along the curve. As k increases, the curve becomes increasingly steep, strengthening the multiplicity adjustment to account for the growing number of models of comparable complexity. Even though w does not appear explicitly in (11), the amount of multiplicity correction under the FB approach still depends on the data through the marginal likelihoods, via the posterior model probabilities in (6). Curiously, this prior yields the same prior mean model size ($k/2$) as the “pseudo-objective” choice of $w = 1/2$, even though probability is allocated fundamentally differently among models.

The use of the EB approach can be motivated by the objective of most accurately estimating the true proportion of nonzero covariates in the underlying true model, $\mathcal{M}_T$ (Carlin & Louis, 2000; Scott & Berger, 2010). However, Scott and Berger (2010) argued that this advantage does not outweigh the drawbacks of ignoring the uncertainty associated with estimating w . In particular, their simulation study showed that, although the EB approach achieved a lower mean estimation error (measured using the ℓ_1 norm) than the FB approach, its poorer performance in terms of the median error was more concerning. This behavior stems from the fact that the EB estimator may not infrequently yield degenerate solutions. For example, $\hat{w}$ is estimated to be exactly 0 whenever the null model has the largest marginal likelihood, regardless of how slight its advantage over the competing models may be. Consequently, the procedure can assign unwarranted certainty to the null model. For these reasons, this work adopts a FB approach for w .

2.5 Exploring Large Model Spaces

In all Bayesian procedures, the posterior distribution quantifies uncertainty after observing the data, encapsulating all information relevant for inference. Consequently, the Bayesian hierarchical mixture approach presented in Section 2.4 can be viewed as a principled framework for exploiting posterior information. Decisions regarding model selection are therefore based on the posterior distribution over models, $f(\boldsymbol{\gamma} \mid \mathbf{Y})$. They can be formally justified from a decision-theoretic perspective, with optimal decisions being intrinsically linked to the posterior predictive distribution of a unknown of interest. See M. A. Clyde and George (2004) for details.

In this work, focus will be given to the Highest Probability Model (HPM), also known as maximum *a posteriori* model (MAP): $\arg \max \{f(\boldsymbol{\gamma} \mid \mathbf{Y}) : \boldsymbol{\gamma} \in \mathcal{M}\}$; and the (marginal) PIPs of predictors, which are obtained by adding up the posterior model probabilities from each model containing $\mathbf{x}_j$, that is, where $\gamma_j = 1$:

$$f(\gamma_j = 1 \mid \mathbf{Y}) = \sum_{\boldsymbol{\gamma}: \gamma_j=1} f(\boldsymbol{\gamma} \mid \mathbf{Y}), \quad j = 1, \dots, k. \quad (12)$$

Both HPM and PIPs are deterministic functions of (6). Insofar as $f(\boldsymbol{\gamma})$ provide an initial representation of model uncertainty, $f(\boldsymbol{\gamma} \mid \mathbf{Y})$ provides a complete representation of postdata model uncertainty. For example, selection of the most probable model in light of the data corresponds to maximizing the expected 0–1 utility for a correct selection. PIPs provide a complementary summary of $f(\boldsymbol{\gamma} \mid \mathbf{Y})$ and naturally lead to a class of selection rules based on individual variable evidence, especially valuable for a scenario as the one of Chapter 4. They are also particularly useful when $|\mathcal{M}|$ is sufficiently large to make overall single-model posterior probabilities substantially low and therefore essentially uninformative for inference. Apart from this, they allow for the construction of the Median Probability Model (MPM) of Barbieri and Berger (2004), defined as the model that includes all variables with PIPs exceeding the 1/2 threshold. Under a variety of settings considered therein and later generalized by Barbieri et al. (2021), this selection strategy exhibits strong, and in some cases optimal, predictive properties under mean-squared predictive loss. This model often differs from the HPM, outperforming it.

In high-dimensional settings, however, exact evaluation of posterior model probabilities in (6) becomes infeasible and posterior summaries can no longer be obtained exactly. The main difficulty lies in the computation of the normalizing constant, C , since its exact value is virtually unknown, due to the enormous size of $\mathcal{M}$, inhibiting exact independent sampling from the posterior distribution. Indeed, the number of possible models $|\mathcal{M}| = 2^k$, grows exponentially with the number of candidate predictors k . Representing the model space in binary form alone illustrates its computational burden (García-Donato & Martínez-Beneito, 2013). This highlights the computational infeasibility of brute-force approaches as k grows, even before considering the cost of evaluating each model. This infeasibility motivated the development of two sampling approaches to efficiently explore the entire model space by selecting only a subset of models (hereafter denoted by $\mathcal{M}^*$), whose size

is small relative to $|\mathcal{M}|$. The quality of the estimators of posterior summaries depends heavily on the sampling mechanism used to obtain $\mathcal{M}^*$, which may or may not contain replicated models.

The first sampling mechanism is closely related to the *Hansen–Hurwitz* estimator (Thompson, 2002), an unbiased estimator under unequal-probability sampling with replacement from a finite population when based on independent draws. Naturally, this does not preclude the existence of repeated models in $\mathcal{M}^*$. Since the goal is to estimate $f(\gamma | \mathbf{Y})$, models should be sampled with probabilities proportional to this quantity. From (6), this corresponds to a probability-proportional-to-size sampling design, where the size measure is $f(\mathbf{Y} | \gamma) f(\gamma)$ (Lohr, 1999, Ch. 6). Generating independent draws from $f(\gamma | \mathbf{Y})$ over huge model spaces is, however, challenging, making traditional Monte Carlo methods inadequate (Hoff, 2009). This stems from the fact that the sampling frame, which specifies all “size” variables, is of the same computational complexity as enumerating the model space. As a result, the standard practice is to obtain dependent samples with the aid of MCMC algorithms.

The key idea behind MCMC algorithms (George & McCulloch, 1997; Hastings, 1970; Raftery, Madigan, & Hoeting, 1997) is that they simulate a sequentially dependent sample,

$$\gamma^{(0)}, \gamma^{(1)}, \gamma^{(2)}, \dots, \gamma^{(S)} \quad (S \geq |\mathcal{M}^*|), \quad (13)$$

whose distribution converges to the stationary distribution, namely $f(\gamma | \mathbf{Y})$. The choice of the initial state of the chain, $\gamma^{(0)}$, influences only the rate of convergence and not the stationary distribution. This result is formalized by the ergodic theorem (Hoff, 2009). It requires that the sequence in (13) satisfy the rather lenient conditions of forming an irreducible, positive recurrent, and aperiodic Markov chain. A Markov chain is a stochastic process satisfying the *markovian* property, whereby the distribution of $\gamma^{(s)}$ depends on the past only through its immediate predecessor $\gamma^{(s-1)}$. Irreducibility, positive recurrence, and aperiodicity are additional conditions ensuring convergence to the stationary distribution. See also Robert and Casella (2004, Ch. 6) for a precise treatment of these concepts. With the additional assumption that the state space $\mathcal{M}$ is finite, García-Donato and Martínez-Beneito (2013) show that *Hansen–Hurwitz*(-type) estimators based on an MCMC sample drawn from $f(\gamma | \mathbf{Y})$ are consistent and, consequently, asymptotically unbiased (i.e., unbiased in the limit as $S \rightarrow \infty$). This essentially means that posterior model probabilities can be consistently estimated by the relative Monte Carlo frequencies of sampled models ($\mathcal{M}^*$), and that deterministic functions of (6) are well approximated by their corresponding empirical counterparts. In the case of estimating the probability of a single model $\mathcal{M}_\gamma$, the estimate will just be $\hat{p}^{\text{MC}}(\mathcal{M}_\gamma | \mathbf{Y}) = (1/S) \sum_{s=1}^S I(\gamma = \gamma^{(s)})$. To improve the performance of the underlying MCMC estimators and to help determine an appropriate stopping point for the simulation, an explicit estimate of C can be obtained. One such approach was proposed by George and McCulloch (1997). An exception is the HPM estimator, whose performance depends solely on the sampler’s ability to discover high-probability models, as its computation does not involve C .

In contrast, the second sampling technique, referred to as the *renormalized ap-*

proach by García-Donato and Martínez-Beneito (2013), relies on heuristic procedures to perform stochastic or deterministic sampling, typically without replacement, from posterior distributions. As a result, the sampled models are generally dependent. The aim is to discover the most probable models, as in the stochastic search algorithm of Berger and Molina (2005) or in the Bayesian Adaptive Sampling method of M. A. Clyde, Ghosh, and Littman (2011). Estimation of deterministic functions of $f(\gamma \mid \mathbf{Y})$ are based on a renormalized analytical expression of (6) for the visited models ($\mathcal{M}^*$),

$$\hat{p}^{RM}(\mathcal{M}_\gamma \mid \mathbf{Y}) = \frac{\text{BF}_{\mathcal{M}_\gamma:\mathcal{M}_\emptyset} f(\mathcal{M}_\gamma)}{\sum_{\mathcal{M}_\ell \in \mathcal{M}^*} \text{BF}_{\mathcal{M}_\ell:\mathcal{M}_\emptyset} f(\mathcal{M}_\ell)}. \quad (14)$$

From a theoretical standpoint, these estimators, known in sampling theory as *ratio* estimators (Thompson, 2002), tend to be very precise (i.e., have a small mean squared error; Casella and Berger, 2002, Ch. 7), but are typically biased. The theoretical justification for sampling without replacement is precisely that the most accurate estimate of (14) is obtained from a sample $\mathcal{M}^*$ consisting of the $|\mathcal{M}^*|$ models with the highest posterior probabilities (Royall, 1970). Additionally, prior to the work of García-Donato and Martínez-Beneito (2013), it was widely believed that, in large model spaces, any given model was unlikely to be visited sufficiently often to yield a reliable Monte Carlo estimate of its posterior probability (Berger & Molina, 2005).

However, the experimental results reported by García-Donato and Martínez-Beneito (2013) indicate that the empirical approach produces fairly accurate and highly precise estimates for PIPs, confirming the high efficiency of such estimators. Moreover, it substantially outperforms renormalized methods, which exhibit clear bias and high variability, a consequence of PIPs being strongly influenced by the estimated HPM in this setting. These differences become more pronounced as the size of $\mathcal{M}$ increases, with the quality of empirical estimates remaining largely unaffected, in contrast to renormalized estimates. Thus, for the empirical approach, the sample size $\mathcal{M}^*$ needed to reach certain precision tends to finite values, even as $|\mathcal{M}|$ becomes arbitrarily large. Furthermore, García-Donato and Martínez-Beneito (2013) show that the empirical approach also has at least as good an ability to identify high-probability models as the renormalized approach, despite the latter being specifically designed for this purpose. The authors, however, caution that while any MCMC method using *empirical* estimators has the potential to produce reliable results, this depends on the specific characteristics of the MCMC algorithm.

2.5.1 Gibbs Sampler

The results of García-Donato and Martínez-Beneito (2013) rely on the assumption of a finite state space, which is one of the conditions ensuring the consistency of the ergodic averages. This assumption is only satisfied when a closed-form expression for (7) is available. In the present work, these results are nevertheless used, even though closed-form expressions are derived only for the approximate marginal like-

lihoods through integrated Laplace approximations (Wang & George, 2007). More details are provided in the next chapter. Accordingly, the practical experiments presented in this work employ the random-order componentwise Gibbs sampler (George & McCulloch, 1997), the MCMC sampling algorithm representative of the *empirical* approach in García-Donato and Martínez-Beneito (2013).

The motivation for the Gibbs Sampler (Gelfand & Smith, 1990) stems from the difficulty of sampling directly from the joint posterior distribution over the model space $f(\boldsymbol{\gamma} \mid \mathbf{Y})$. In general, it is a non-standard multidimensional distribution, i.e., it cannot be expressed in terms of well-known probability distributions for which direct sampling algorithms are available (see, e.g., Ross, 2006, Chs. 3–5). In contrast, sampling from the (uni-dimensional) full conditional distribution of each γ_j ,

$$f(\gamma_j \mid \mathbf{Y}, \boldsymbol{\gamma}_{(j)}) = \frac{f(\boldsymbol{\gamma} \mid \mathbf{Y})}{f(\boldsymbol{\gamma}_{(j)} \mid \mathbf{Y})} \propto f(\boldsymbol{\gamma} \mid \mathbf{Y}), \quad \boldsymbol{\gamma}_{(j)} = (\gamma_i : i \neq j), \quad (15)$$

is much easier, fast and efficient. Since γ_j is a binary random variable, its full conditional distribution is Bernoulli, and to fully specify it, it suffices to compute

$$f(\gamma_j = 1 \mid \mathbf{Y}, \boldsymbol{\gamma}_{(j)}) = \frac{f(\gamma_j = 1 \mid \mathbf{Y}, \boldsymbol{\gamma}_{(j)})}{f(\gamma_j = 1 \mid \mathbf{Y}, \boldsymbol{\gamma}_{(j)}) + f(\gamma_j = 0 \mid \mathbf{Y}, \boldsymbol{\gamma}_{(j)})} = \frac{o_j}{o_j + 1}, \quad (16)$$

where o_j denotes the conditional (posterior) odds that γ_j is 1, given by

$$\frac{f(\gamma_j = 1 \mid \mathbf{Y}, \boldsymbol{\gamma}_{(j)})}{f(\gamma_j = 0 \mid \mathbf{Y}, \boldsymbol{\gamma}_{(j)})} = \frac{f(\mathbf{Y} \mid \gamma_j = 1, \boldsymbol{\gamma}_{(j)})}{f(\mathbf{Y} \mid \gamma_j = 0, \boldsymbol{\gamma}_{(j)})} \times \frac{f(\gamma_j = 1, \boldsymbol{\gamma}_{(j)})}{f(\gamma_j = 0, \boldsymbol{\gamma}_{(j)})}. \quad (17)$$

This last equality follows directly from (6), as the normalizing constant C cancels in the ratio. The random-order componentwise variant of the Gibbs sampler generates the sequence in (13) as presented on Page 167 of Hoff (2009). Note that many of the visited models are intermediate states and are therefore not used in the computation of the empirical frequencies. Additionally, when drawing from (15), the sampled value of γ_j may coincide with its current value, resulting in a self-transition for that component update.

3 Model Selection with Factors

Consider the equation introduced in (5). Hereafter, numerical explanatory variables are indexed by i rather than j , while observations are indexed by t . Thus, a numerical variable $\mathbf{x}_i$ is considered active if its associated coefficient β_i differs from zero. However, when the pool of potential predictors also includes factors (i.e., categorical variables), additional notation and care are required. Throughout this dissertation, a factor $\mathbf{\Lambda}_j$ is considered active, or relevant, if at least one of the coefficients, λ_{jr} , $r = 1, \dots, \ell_j$, associated with its levels is non-zero. This flexible definition of factor relevance has also been adopted in previous works (see 2016, 1996, 1998, 2002, 2022b). In contrast, other references (2009, 2006) assume that a factor is included in a model only when all of its levels are simultaneously present. However, this assumption may be restrictive. For instance, when a factor is obtained by discretizing a numerical variable, the somewhat arbitrary construction of its levels is not exempt from the complexity–fit trade-off (P. K. Dunn & Smyth, 2018, Ch. 1) and, as a consequence, the relevance of a factor depends on its granularity and, more specifically, on the proportion of active levels to its dimension. Therefore, to properly account for this situation, it is pertinent to consider models that allow for every possible combination of active levels within each factor, rather than restricting attention only to models where factors are either entirely included or entirely excluded.

3.1 Homoskedastic Linear Regression

The set of potential explanatory variables is now composed of $\mathbf{x}_1, \dots, \mathbf{x}_k$ (numerical variables) and factors (categorical variables) $\mathbf{\Lambda}_1, \dots, \mathbf{\Lambda}_p$, each with a number of levels denoted by $\ell_1, \dots, \ell_p$ such that $\ell_j \geq 2$, $j = 1, \dots, p$; also, let $L = \sum_{j=1}^p \ell_j$. The full model can then be defined as follows:

$$\mathbf{y} = \mathbf{1}_n \alpha + \mathbf{X} \boldsymbol{\beta} + \mathbf{Z} \boldsymbol{\lambda} + \boldsymbol{\varepsilon}, \quad (18)$$

where the n -dimensional vector of random errors $\boldsymbol{\varepsilon}$ follows a multivariate normal distribution with mean $\mathbf{0}$ and covariance matrix $\sigma^2 \mathbf{I}_n$, with $\mathbf{I}_n$ representing the identity matrix of order n ($w_t = 1, \forall t \in \{1, \dots, n\}$). The parameter vectors are $\boldsymbol{\beta} = (\beta_1, \dots, \beta_k)^\top$, and $\boldsymbol{\lambda} = (\boldsymbol{\lambda}_1, \dots, \boldsymbol{\lambda}_p)^\top$, with $\boldsymbol{\lambda}_j = (\lambda_{j1}, \dots, \lambda_{j\ell_j})^\top$, $j = 1, \dots, p$. Any model in $\mathcal{M}$ (to be defined later) can be identified with an indicator vector $(\boldsymbol{\gamma}, \boldsymbol{\delta}) \in \{0, 1\}^{k+L}$, where $\boldsymbol{\gamma}^\top = (\gamma_1, \dots, \gamma_k)$, $\boldsymbol{\delta}^\top = (\boldsymbol{\delta}_1^\top, \dots, \boldsymbol{\delta}_p^\top)$, $\boldsymbol{\delta}_j^\top = (\delta_{j1}, \dots, \delta_{j\ell_j})$; $\gamma_i = 1$ if $\mathbf{x}_i$ is in the model (0 otherwise) and $\delta_{jr} = 1$ if level r of factor $\mathbf{\Lambda}_j$ is in the model (0 otherwise), $i = 1, \dots, k$, $r = 1, \dots, \ell_j$, $j = 1, \dots, p$. To indicate which factors are active in a particular model $\mathcal{M}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}$, the indicator vector $\boldsymbol{\tau} = \boldsymbol{\tau}(\boldsymbol{\delta}) = (\tau_1(\boldsymbol{\delta}_1), \dots, \tau_p(\boldsymbol{\delta}_p))^\top$ is defined, where each τ_j is a deterministic function of $\boldsymbol{\delta}_j$, taking the value 1 if $\mathbf{1}_{\ell_j}^\top \boldsymbol{\delta}_j \geq 1$ (0 otherwise). $\mathbf{Z} = [\mathbf{Z}_1 \mid \dots \mid \mathbf{Z}_p]$ is a $n \times L$ matrix, where $\mathbf{Z}_j = (z_{tr}^{(j)})$ is an $n \times \ell_j$ matrix, with $z_{tr}^{(j)} = 1$ if the t -th observation takes the r th level of $\mathbf{\Lambda}_j$, and 0 otherwise, $t = 1, \dots, n$, $r = 1, \dots, \ell_j$, $j = 1, \dots, p$. Stating that level r of factor $\mathbf{\Lambda}_j$ is in the model is equivalent to saying that the corresponding column of $\mathbf{Z}_j$ was not deleted. The matrices $\mathbf{X}_\gamma$ and $\mathbf{Z}_\delta$, of dimensions $n \times k_\gamma$ and $n \times p_\delta$ (with $p_\delta = \sum_{j=1}^p \sum_{r=1}^{\ell_j} \delta_{jr}$), respectively, are obtained by selecting the columns of $\mathbf{X}$ and

$\mathbf{Z}$ indexed by γ and δ ; the corresponding subvectors of regression coefficients are denoted by β_γ and λ_δ .

The $z_{\cdot r}^{(j)}$ are called artificial or dummy variables. This terminology assumes the use of treatment coding (P. K. Dunn & Smyth, 2018, Ch. 1), although several alternative coding schemes are available for representing categorical variables numerically in statistical models (see UCLA Statistical Methods and Data Analytics, 2025). Regardless of the coding scheme imposed on factors, least squares estimation (P. K. Dunn & Smyth, 2018, Ch. 2) requires the design matrix $[\mathbf{1}_n \mid \mathbf{X} \mid \mathbf{Z}]$ to be of full rank to ensure the existence of a unique solution. The same requirement applies to maximum likelihood estimation for non-normal EDMs using the Iterative Re-Weighted Least Squares (IRWLS) algorithm (P. K. Dunn & Smyth, 2018, Ch. 6). For treatment coding this translates into choosing a reference level $r(j)$ for each factor Λ_j , setting $\lambda_{j,r(j)} = 0$ and omitting the corresponding column from $\mathbf{Z}_j$, yielding the reduced matrix $\mathbf{Z}_j^*$, for $j = 1, \dots, p$. Then, the effect of each remaining level on the linear predictor (i.e., the mean response in linear regression), *ceteris paribus*, is interpreted as the deviation to the selected reference.

Although the non-identifiable formulation of the variable selection problem in (18) may initially seem like an odd choice, its motivation becomes clearer in light of the definition of factor relevance introduced at the beginning of this chapter. As first noted in Section 5.1 of Fernández, Ley, and Steel (2002) and later investigated more thoroughly by García-Donato and Paulo (2022b), generating the class of competing models by deleting columns from the matrix $[\mathbf{X} \mid \mathbf{Z}^*]$, where $\mathbf{Z}^* = [\mathbf{Z}_1^* \mid \dots \mid \mathbf{Z}_p^*]$, does not yield an exhaustive model space, since not all possible combinations of active factor levels are represented. In particular, the number of entertained models under a full-rank parameterization of the full model in (18) is $2^k \prod_{j=1}^p 2^{\ell_j - 1}$. Additionally, the chosen parameterization determines which models are absent. In the case of treatment coding, the omitted models depend on the particular dummy encoding adopted for the factors, that is, on the choice of the reference levels (see Section S.1 of García-Donato and Paulo, 2022a). This issue is of even greater importance when relevant models, such as the true data-generating process $\mathcal{M}_T$, are *a priori* excluded from the candidate model set, thereby potentially affecting model selection results substantially, independently of the sample size. It is further exacerbated as the number of predictors, $k + p$, increases.

To circumvent this problem and include the missing models, the “free reference solution” proposed by Fernández, Ley, and Steel (2002) instead consists of deleting columns from the $[\mathbf{X} \mid \mathbf{Z}]$ matrix, thereby generating an exhaustive class of competing models. However, the resulting class, of size 2^{k+L} , contains duplicate models, in the sense that they are merely different parameterizations of the same statistical model. Specifically, duplicate models arise whenever, for at least one factor with ℓ levels, either $\ell - 1$ or all of its levels are active. The solution presented by García-Donato and Paulo (2022b) was to remove all but one model from each set of duplicates, so that $\mathcal{M}$ forms an exhaustive set of unique models, of size $|\mathcal{M}| = 2^k \prod_{j=1}^p (2^{\ell_j} - \ell_j)$. As argued by García-Donato and Paulo (2022b), the choice of which representative to retain from each set of duplicate models is itself independent of the reference levels chosen for model fitting, provided that the

model evaluation criterion (e.g., the null-based Bayes Factor in (8)) is invariant under full-rank parameterizations. Their strategy was to retain the overparameterized version, that is, the one with all ℓ levels active. In particular, under conventional priors (see Section 2.4), the above condition holds, since conventional (null-based) Bayes Factors for linear regression depends only on the Residual Sum of Squares (P. K. Dunn & Smyth, 2018, Ch. 2) and the rank of $[\mathbf{1} \mid \mathbf{X}_\gamma \mid \mathbf{Z}_\delta]$. Details can be found in Propositions 1 and 3 of Bayarri and García-Donato (2007).

3.1.1 Multiplicity Correction

From an objective perspective, prior indifference regarding the inclusion of a given variable is typically expressed by assigning it an *a priori* inclusion probability of $1/2$, irrespective of the set of potential explanatory variables under consideration. Similar to (12), for a numerical covariate $\mathbf{x}_i$ and a factor Λ_j , these probabilities are given by

$$f(\gamma_i = 1) = \sum_{\gamma, \delta: \gamma_i=1} f(\gamma, \delta), \quad (19)$$

and

$$f(\tau_j = 1) = \sum_{\gamma, \delta: \tau_j=1} f(\gamma, \delta), \quad (20)$$

where $f(\tau_j = 1)$ is the sum of the prior probabilities of models including at least one level of factor Λ_j . The model space priors presented in Section 2.4 do not account for the grouped structure of categorical variables. Specifically, this representation induces dependence among the entries of the inclusion indicator vector (γ, δ) , thereby violating the assumption of exchangeable Bernoulli trials and, when $\ell_j > 2$, the indifference principle with respect to $f(\tau_j = 1)$. Extensions these priors to accommodate categorical variables are presented in Section 3.2. of García-Donato and Paulo (2022b). However, as shown in Section S.4 of García-Donato and Paulo (2022a), these priors exhibit several undesirable properties. In particular, $f(\tau_j = 1)$ increases with ℓ_j , regardless of the relevance of the additional levels. While this behavior is expected under the constant prior, it is somewhat surprising under the SB prior. An equally problematic feature these priors is that $f(\tau_j = 1)$ does not decrease as the number of competing variables, $k + p$, increases. Recall that Scott and Berger (2010) warned that multiplicity correction should arise through the prior probabilities assigned to models, as discussed in Section 2.4. It is also important to note that the source of this problem differs from that arising when models are generated from either the overparameterized full model (18) or its full-rank parameterization. For the latter, H. A. Chipman (1996) emphasized the need to control for multiplicity within the group of dummy variables defining a factor. Nonetheless, this recommendation was largely overlooked in the literature until the solution proposed by García-Donato and Paulo (2022b) for the overparameterized formulation.

Their solution addresses this issue by constructing the model space prior hierarchically, in two stages, as

$$f(\gamma, \delta) = f(\gamma, \delta, \tau) = f(\delta \mid \gamma, \tau) f(\gamma, \tau), \quad (21)$$

where the “marginal” prior $f(\boldsymbol{\gamma}, \boldsymbol{\tau})$ is designed to control the prior inclusion probabilities $f(\gamma_i = 1)$ and $f(\tau_j = 1)$. In particular, under the constant and the SB-type priors (see section 3.2 of García-Donato and Paulo, 2022b), these inclusion probabilities are fixed at $1/2$, regardless of the number of candidate variables or the number of levels defining each factor. However, only the SB-type prior accounts for the multiplicity associated with the total number of candidate predictors, $k + p$, analogously to the setting of Scott and Berger (2010). The second stage of the model space prior in (21), namely the “conditional” prior $f(\boldsymbol{\delta} \mid \boldsymbol{\gamma}, \boldsymbol{\tau})$, distinguishes the probability assigned to models with the same combination of active factors and numerical variables, i.e., models belonging to the set $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau}) = \{(\boldsymbol{\gamma}, \boldsymbol{\delta}) \in \mathcal{M} : \boldsymbol{\gamma} = \boldsymbol{\gamma}, \min\{\mathbf{1}_{\ell_j}^\top \boldsymbol{\delta}_j, 1\} = \tau_j, j = 1, \dots, p\}$. Two “objective” specifications for this conditional prior were investigated by García-Donato and Paulo (2022b) - see Section 3.2. for their explicit expression: an equally distribution of probability mass across the different model ranks of the models in $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})$, $\text{rank}([\mathbf{1}_n \mid \mathbf{X}_\gamma \mid \mathbf{Z}_\delta])$ (SB-type); and a distribution of probability mass across model ranks directly proportional to the number of models in $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})$ with the same model rank (Constant-type). Note that, since there are 2^{k+p} possible combinations of $(\boldsymbol{\gamma}, \boldsymbol{\tau})$, there are also 2^{k+p} corresponding sets of models $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})$ arises. The cardinality of each set $|\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})|$ is derived in Section S.4 of García-Donato and Paulo (2022a).

The authors recommend adopting SB-type priors as the default specification for both the marginal and conditional priors. However, the motivation for the latter choice is not to control multiplicity as the numbers of factor levels, ℓ_j , increase, since both of the proposed conditional priors are surprisingly sensitive to this phenomenon. Instead, their choice is related to the average of the null-based Bayes Factor (8) computed over the models in $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})$ with the same model rank, $r = \text{rank}([\mathbf{1}_n \mid \mathbf{X}_\gamma \mid \mathbf{Z}_\delta])$. This average is denoted by $\overline{\mathcal{B}}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}^r$ throughout this work. Then, as argued by García-Donato and Paulo (2022b), when adopting a SB-type conditional prior, the evidence that the data assign to a given combination $(\boldsymbol{\gamma}, \boldsymbol{\tau})$, denoted by $\mathcal{B}_{\boldsymbol{\gamma}, \boldsymbol{\tau}}^{\text{SB}}$, can be interpreted as the simple average of $\overline{\mathcal{B}}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}^r$ over all model ranks in $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})$. Consequently, it provides robust evidence in favor of $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})$, with respect to the rank of the true model, $\mathcal{M}_T$. In contrast, under a Constant-type conditional prior, $\mathcal{B}_{\boldsymbol{\gamma}, \boldsymbol{\tau}}^{\text{C}}$ corresponds to a weighted average of $\overline{\mathcal{B}}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}^r$, assigning greater weight to intermediate ranks in $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})$, irrespective of the rank of the true model. These differences are more pronounced for sets of models $\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})$ of higher cardinality.

Beyond the variety of desiderata satisfied by conventional priors (Bayarri et al., 2012), the variable selection methodology developed by García-Donato and Paulo (2022b) offers several additional advantages. Compared with other model selection approaches, it explicitly accounts for multiplicity with respect to the number of competing predictors, $k + p$. In contrast to frequentist estimation methods, it is also simpler to implement, as it does not require the specification of tuning parameters. Furthermore, the frequentist performance of this methodology was shown to compare favourably with that of competing approaches. Motivated by these considerations, the remainder of this chapter is devoted to extending the approach of García-Donato and Paulo (2022b) to the broader class of GLMs.

3.2 g-priors in Generalized Linear Models

Using the notation introduced in (18), the systematic component (5) can be written as

$$\boldsymbol{\eta}_{\gamma,\delta} = \mathbf{1}_n \alpha + \mathbf{X}_\gamma \boldsymbol{\beta}_\gamma + \mathbf{Z}_\delta \boldsymbol{\lambda}_\delta = \mathbf{1}_n \alpha + \tilde{\mathbf{X}}_{\gamma,\delta} \boldsymbol{\beta}_{\gamma,\delta}, \quad (22)$$

where $\tilde{\mathbf{X}}_{\gamma,\delta} = [\mathbf{X}_\gamma \mid \mathbf{Z}_\delta]$ is the design matrix with $k_{\gamma,\delta} = k_\gamma + p_\delta$ columns and $\boldsymbol{\beta}_{\gamma,\delta} = (\boldsymbol{\beta}_\gamma, \boldsymbol{\lambda}_\delta)^\top$. It is assumed throughout that $\tilde{\mathbf{X}} = [\mathbf{X} \mid \mathbf{Z}]$ has full-column rank, with $k + p \leq n$, as its column space does not contain $\mathbf{1}_n$. In particular, this implies that all categories of the factors are observed. Henceforth, attention is restricted to members of the EDM family with unit dispersion parameter ($\phi = 1$), such as the Poisson, Binomial, and Exponential distributions. Consequently, there is no need of taking a FB treatment of ϕ , as in linear regression (see, e.g., Hoff, 2009), or an EB approach, as in Wang and George (2007), and (7) reduces to

$$f(\mathbf{Y} \mid \mathcal{M}_{\gamma,\delta}) = \int_{\mathbb{R}^{k_\gamma + p_\delta}} \int f(\mathbf{Y} \mid \alpha, \boldsymbol{\beta}_{\gamma,\delta}, \mathcal{M}_{\gamma,\delta}) f(\boldsymbol{\beta}_{\gamma,\delta} \mid \mathcal{M}_{\gamma,\delta}) f(\alpha) d\alpha d\boldsymbol{\beta}_{\gamma,\delta}. \quad (23)$$

To overcome the intractability of the integral in (23) (see Section 2.4), Laplace's method is used (Kass & Raftery, 1995). In its simplest formulation, it approximates a well-behaved, unimodal function by a normal distribution. This approach is preferable to standard numerical methods, which may become inefficient for such integrals, particularly in high-dimensional settings where the integrand (a quantity proportional to the posterior density) is highly concentrated around its maximum, namely the posterior mode. This is the case when the likelihood function $f(\mathbf{Y} \mid \alpha, \boldsymbol{\beta}_{\gamma,\delta}, \mathcal{M}_{\gamma,\delta})$ is highly peaked near its maximum, the Maximum Likelihood Estimate (MLE), especially for large samples, owing to the asymptotic equivalence of the two estimators (Robert, 2007, Ch. 4).

An alternative form of the Laplace approximation (Kass, Tierney, & Kadane, 1990), which is more advantageous computationally and has the same order of relative error, $O(n^{-1})$, relies on the MLEs, as these are easily computed by statistical software (e.g., M. Clyde, 2025). It first applies a second-order Taylor series expansion to $\log f(\mathbf{Y} \mid \alpha, \boldsymbol{\beta}_{\gamma,\delta}, \mathcal{M}_{\gamma,\delta})$ around the MLE $\hat{\mathbf{c}} = (\hat{\alpha}_{\gamma,\delta}, \hat{\boldsymbol{\beta}}_{\gamma,\delta})$. After exponentiation, this yields an approximation to the likelihood in the form of the p.d.f. of a multivariate normal distribution with mean $\hat{\mathbf{c}}$ and covariance matrix $\mathcal{J}_n(\hat{\alpha}_{\gamma,\delta}, \hat{\boldsymbol{\beta}}_{\gamma,\delta})^{-1}$, where $\mathcal{J}_n(\hat{\alpha}_{\gamma,\delta}, \hat{\boldsymbol{\beta}}_{\gamma,\delta})$ is the observed Fisher information matrix, i.e., the negative Hessian matrix of the log-likelihood (P. K. Dunn & Smyth, 2018), evaluated at the MLEs $\hat{\mathbf{c}}$. Then, this approximation is substituted, along with $f(\boldsymbol{\beta}_{\gamma,\delta} \mid \mathcal{M}_{\gamma,\delta}) f(\alpha) \approx f(\hat{\boldsymbol{\beta}}_{\gamma,\delta} \mid \mathcal{M}_{\gamma,\delta})$ (explained later), into (23), and the resulting expression is integrated with respect to α and $\boldsymbol{\beta}_{\gamma,\delta}$. However, as noted by Wang and George (2007), this variant performs poorly for the normal EDMs, failing to recover the correct marginal likelihood for certain values of the hyperparameter g . To overcome this limitation, Wang and George (2007) proposed the iL approximation, with the same order of relative error, $O(n^{-1})$. The key difference is that less is approximated, since $f(\boldsymbol{\beta}_{\gamma,\delta} \mid \mathcal{M}_{\gamma,\delta})$ is not replaced by $f(\hat{\boldsymbol{\beta}}_{\gamma,\delta} \mid \mathcal{M}_{\gamma,\delta})$ in (23). Thus,

when the iL approximation is applied to both the numerator and denominator of the Bayes factor in (8), the resulting approximation preserves the same order of relative error.

The implementation of the iL approximation in GLMs, nonetheless, requires modifications to the normal g-priors, such as the one in (9). The reason is that these priors are commonly defined in terms of the expected Fisher information matrix, $\mathcal{I}_n(\beta_{\gamma,\delta}) = \mathbb{E}[\mathcal{J}_n(\beta_{\gamma,\delta})]$, which depends on the parameters α and $\beta_{\gamma,\delta}$, for non-normal EDMs. Consequently, the corresponding covariance matrix is not fixed, and the prior is not properly defined until a rule is supplied for evaluating it.

Two decisions are involved in such a rule. The first is *which* information matrix to use, the expected $\mathcal{I}_n$ (Fouskakis, Ntzoufras, & Draper, 2009b) or the observed $\mathcal{J}_n$ (Wang & George, 2007); the second is *where* to evaluate it, either at the null value $\beta_{\gamma,\delta} = \mathbf{0}$ or at the model-specific MLE $\hat{\beta}_{\gamma,\delta}$. Both decisions are settled here in favour of the observed information evaluated at the MLE, for two reasons. Although classical asymptotic theory presents $\mathcal{I}_n(\beta_{\gamma,\delta})$ as the measure of the large-sample precision of $\hat{\beta}_{\gamma,\delta}$, Efron and Hinkley (1978) recommend $\mathcal{J}_n(\beta_{\gamma,\delta})$ as a more accurate assessment of that same quantity for the data at hand. Moreover, when the true model is not the null one, $\mathcal{M}_T \neq \mathcal{M}_\emptyset$, evaluating the information matrix at $\hat{\beta}_{\gamma,\delta}$ (Wang & George, 2007) captures the large-sample covariance structure of $\beta_{\gamma,\delta}$, and the local geometry of the likelihood under $\mathcal{M}_{\gamma,\delta}$, more faithfully than an anchor at a null value that may lie far from the region where the likelihood concentrates. As an additional benefit, the local orthogonality that this choice induces between α and $\beta_{\gamma,\delta}$ simplifies the posterior derivations that follow and reduces the computational burden of exploring the model space. This led Li and Clyde (2018) to propose the Local Information Metric (LIM) g-prior, introduced next:

$$\beta_{\gamma,\delta} \mid \mathcal{M}_{\gamma,\delta}, g \sim \mathcal{N}(\mathbf{0}, g \cdot \mathcal{J}_n(\hat{\beta}_{\gamma,\delta})^{-1}). \quad (24)$$

As hinted in Section 2.4, the parameter common to every model is assigned an improper prior of the type

$$f(\alpha) \propto 1. \quad (25)$$

Historically, the use of (25) was justified by first recentering the columns of the design matrix to make them orthogonal to $\mathbf{1}_n$. However, Bayarri et al. (2012) noted that this orthogonalization is unnecessary. Instead, they justify the improper prior through two key properties of the g-prior: predictive matching and location invariance, both of which are satisfied by the LIM g-prior in (24). In particular, integrating the marginal likelihood in (23) with respect to the improper uniform measure on α yields a marginal likelihood that is invariant to arbitrary location shifts in the columns of $\mathbf{X}$. Furthermore, under the prior in (25), the marginal posterior distribution of α is asymptotically proper and, for large samples, asymptotically independent of the posterior distribution of $\beta_{\gamma,\delta}$. The latter property follows from the local orthogonality between α and $\beta_{\gamma,\delta}$. As illustrated in Section 4.2, this asymptotic independence has important practical implications for posterior inference.

3.2.1 Approximate Bayes Factor

Under an iL approximation, the LIM g-prior in (24) yields closed-form expressions for the integral in (23) and, thus, for the null-based Bayes factors in (8) (Li & Clyde, 2018), that depend on quantities related to MLEs (P. K. Dunn & Smyth, 2018):

$$\text{BF}_{\mathcal{M}_{\gamma,\delta}:\mathcal{M}_{\emptyset}} = \left[\frac{\mathcal{J}_n(\hat{\alpha}_{\mathcal{M}_{\emptyset}})}{\mathcal{J}_n(\hat{\alpha}_{\mathcal{M}_{\gamma,\delta}})} \right]^{\frac{1}{2}} (1+g)^{-\frac{k_{\gamma,\delta}}{2}} \exp \left\{ \frac{z_{\mathcal{M}_{\gamma,\delta}}}{2} - \frac{Q_{\mathcal{M}_{\gamma,\delta}}}{2(1+g)} \right\}, \quad (26)$$

where

$$Q_{\mathcal{M}} = \hat{\beta}_{\gamma,\delta}^{\top} \mathcal{J}_n(\hat{\beta}_{\gamma,\delta}) \hat{\beta}_{\gamma,\delta} \quad (27)$$

is the Wald test statistic under observed information (P. K. Dunn & Smyth, 2018) and

$$z_{\mathcal{M}_{\gamma,\delta}} = 2 \log \left\{ \frac{f(\mathbf{Y} \mid \hat{\eta}_{\mathcal{M}_{\gamma,\delta}}, \mathcal{M}_{\gamma,\delta})}{f(\mathbf{Y} \mid \hat{\eta}_{\mathcal{M}_{\emptyset}}, \mathcal{M}_{\emptyset})} \right\} \quad (28)$$

is the observed likelihood ratio test statistic, when comparing model $\mathcal{M}$ with the null model $\mathcal{M}_{\emptyset}$ (P. K. Dunn & Smyth, 2018). The expression in (26) is consistent with the Bayesian Ockham’s razor principle (Jefferys & Berger, 1992), as it naturally balances model complexity against goodness of fit. Model complexity is penalized through the term $(1+g)^{-k_{\gamma,\delta}/2}$, while the exponential term rewards better models that provide a better fit to the data. An analogous interpretation applies to the approximate (null-based) Bayes Factor under the Confluent Hypergeometric Information Criterion (CHIC) g-prior (see Section 3.1 of Li and Clyde, 2018). More details on the observed Fisher information matrices and the derivation of this expression are provided in the Supplementary Material.

The model space considerations proposed by García-Donato and Paulo (2022b) are based on the premise that duplicate models receive the same value under the model selection criterion. In Appendix A it is shown that the approximate (obtained via iL approximations) null-based Bayes factor for the LIM g-prior in (26), as well as the approximate Bayes factor for the CHIC g-prior in Section 3.1 of Li and Clyde (2018), possess the same invariance property: they are unchanged by the choice of reference level for categorical variables in duplicate models. This result indicates that the methodology of García-Donato and Paulo (2022b) can be naturally extended to GLMs. The following simulation studies examine whether the resulting priors retain their multiplicity correction properties in this broader setting.

3.3 Simulation Experiments

This section revisits Experiments 1 and 2 from García-Donato and Paulo (2022b), with slight variations, to examine the effect of different model prior specifications on multiplicity control, by investigating their impact on the posterior distribution. As in García-Donato and Paulo (2022b), the prior on the model parameters is the Robust prior with the recommended hyperparameter values proposed by Bayarri et al. (2012). This choice is consistent with the recommendations from the empirical

studies of Li and Clyde (2018). The results presented are from a Poisson EDM with its canonical link function, $\eta = \log(\mu)$, although other EDMs have also been tested (see Chapter 5 for more details on this). In both experiments, $n = 300$ observations for the linear predictor are drawn based on the right-hand side of (18) and the way the design matrix was sampled did not change to the previous experiment. The intercept was set to $\alpha = 1$. However, given that in log-linear models (P. K. Dunn & Smyth, 2018) the effect of the coefficients λ_{jr} are interpreted multiplicatively on the mean μ_t , rather than additively as in the normal EDM with the identity link function, changes were made to the coefficients used in the experiments of García-Donato and Paulo (2022b).

3.3.1 Choice of First-Stage Priors

In the first experiment, similar in spirit to the experiment of Scott and Berger (2010), the focus is on studying the behavior of the marginal inclusion probabilities of the entertained variables, p factors with $\ell = 4$ levels each, as spurious variables are included, without changing the true model. The true model is here considered to consist of two active factors, $\mathbf{\Lambda}_1$ and $\mathbf{\Lambda}_2$, with only active levels. Three scenarios were considered, with an increasing number of spurious factors $p - 2 \in \{2, 13, 25\}$, meaning $\lambda_j = 0$, for $3 \leq j \leq p$. Each of these scenarios was tested 10 times, with different design matrices and different randomly chosen initial models for the Gibbs sampler.

For the active factors, it is important to note that the percentage increase or decrease in the expected response μ_t when δ_{jr} turns from 0 to 1, *ceteris paribus*, is related to $e^{\lambda_{jr}}$, although its actual effect depends on which model is being tested (i.e., which other levels of the same factor are included), and in the case of over-parameterized models, depends on the actual values of the coefficients chosen for the other levels of the factors. This means defining an unambiguously active factor for simulation studies with factors is not a straightforward task. The choice was to shrink all the coefficients in $\mathbf{\Lambda}_1$ were shrunk by 50% relative to the previous experiments, while for $\mathbf{\Lambda}_2$ this transformation was applied only to the coefficient corresponding to the first level:

$$\mathbf{\lambda}_1^\top = (-0.54, -0.42, -0.37, 0.315), \quad \mathbf{\lambda}_2^\top = (-0.255, 0.41, 0.18, 0.07).$$

Figure 1 displays the results of the PIPs under the three prior specifications. $S = 20,000$ iterations of the Gibbs sampler were performed after a burn-in period of $B = 2,000$ iterations. Given that the number of marginal likelihood evaluations per iteration increases with $L = \ell \times p$, and that the cost of evaluating a model is still higher than in linear regression, the chain was stopped once the number of sampled distinct models, $|\mathcal{M}^*|$, surpassed 5000. The results are consistent with those in García-Donato and Paulo (2022b). The increase in the number of spurious factors has little to no effect on the number of false positives (red points) for model priors with an SB-type marginal, never exceeding 2 in any case, providing further evidence that these priors control multiplicity with respect to the inclusion of factors, even

outside the linear regression framework. In contrast, adopting a constant prior for the first stage has a negative effect, with the number of false signals corresponding to spurious factors increasing with p : 0, 6, and 19 for $p = 5$, $p = 15$, and $p = 27$, respectively.

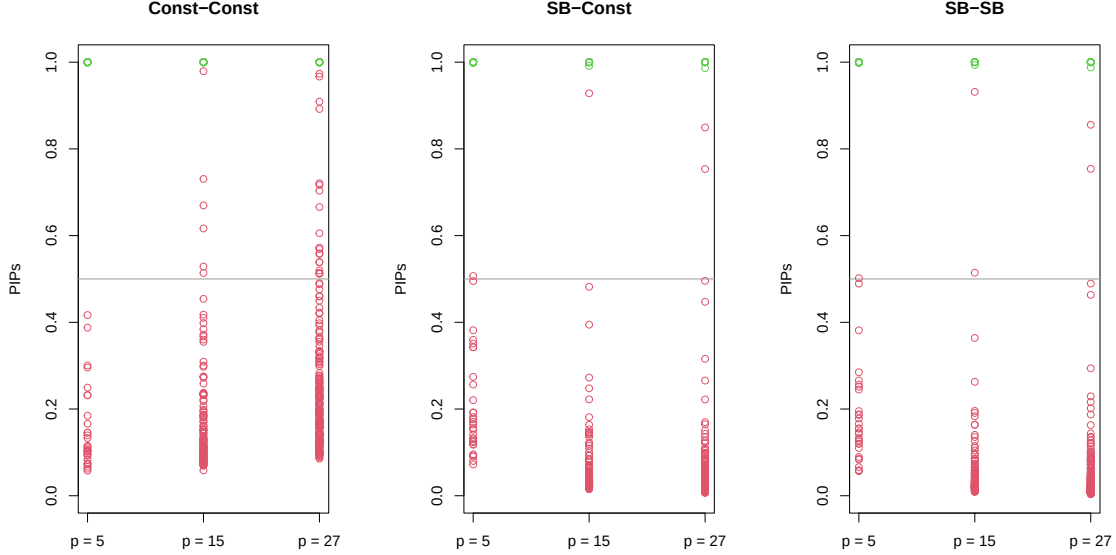


Figure 1: PIPs for Experiment 1 under the three prior specifications

3.3.2 Comparing Second-Stage Prior Choices

Experiment 2 examines the effect on the PIPs of increasing the number of spurious categories within a fixed set of competing factors ($p = 4$), all with the same number of levels ℓ , with $\ell = 5$ or $\ell = 15$. The goal is to determine the most appropriate type of conditional prior to adopt, restricting attention, based on the results of Experiment 1, to the specifications with an SB-type marginal. The setting is one in which all model-specific coefficients are null, apart from the one corresponding to the first level of the first factor, i.e., $\lambda_{11} \neq 0$. The experiment was repeated 10 times, using different seeds, and two different values for this coefficient, $\lambda_{11} = 0.55$ and $\lambda_{11} = 0.70$. Given that only one level of the factor is active, this corresponds, respectively, to a percentage increase of 70% and 100% in the response means for the observations from this category compared to the other observations. More considerations about the choice of the model parameters are given in Chapter 5. Given that this experiment is computationally less expensive than Experiment 1, in terms of memory, no restriction on the size of $\mathcal{M}^*$ was imposed, and the Gibbs samplers were run with $S = 100,000$ and $B = 1,000$. The results in Table 1 correspond to the average PIPs across the 10 simulation replicates.

The results are somewhat different from those obtained in Table 1 of García-Donato and Paulo (2022b). In particular, for the lower value of ℓ , one would have expected closer agreement between the two prior options. Perhaps the most striking differences from the experiment in García-Donato and Paulo (2022b) are the ability

	$\lambda_{11} = 0.70$				$\lambda_{11} = 0.55$			
	SBSB		SBC		SBSB		SBC	
	$\ell = 5$	$\ell = 15$	$\ell = 5$	$\ell = 15$	$\ell = 5$	$\ell = 15$	$\ell = 5$	$\ell = 15$
$f(\tau_1 = 1 \mid \mathbf{Y})$	1.00	1.00	1.00	1.00	1.00	0.90	0.87	0.62
$f(\tau_2 = 1 \mid \mathbf{Y})$	0.15	0.08	0.09	0.00	0.07	0.08	0.03	0.00
$f(\tau_3 = 1 \mid \mathbf{Y})$	0.12	0.07	0.07	0.00	0.09	0.07	0.07	0.00
$f(\tau_4 = 1 \mid \mathbf{Y})$	0.07	0.07	0.03	0.00	0.11	0.08	0.02	0.00

Table 1: Mean PIPs across 10 simulation replicates in Experiment 2

of the SBSB prior to detect the active factor more consistently than the SBC prior as the value of λ_{11} decreases; and the decrease in the average PIPs of the spurious factors under SBSB as ℓ increases. On the other hand, an intriguing aspect is the lower PIPs attributed for spurious factors under SBSB for the lower value of λ_{11} , although this is not problematic given the relatively small differences. Values of λ_{11} above 0.70 were also tested, but the corresponding results are omitted, as there was a reasonably good agreement between both choices, especially in terms of the inclusion of the active factor. As for the results in Table 1, they were volatile across repetitions, especially for the scenarios with $\ell = 15$ and the lower value of λ_{11} . This further illustrates the difficulty of finding appropriate values of λ_{11} for conducting a simulation experiment comparable to that of García-Donato and Paulo (2022b) for linear regression.

While the volatility of the results prevents drawing strong conclusions regarding multiplicity control, the evidence presented here, together with the arguments in Section 3.1, supports the recommendation of SB-type priors for both stages. Nonetheless, the next chapter presents a scenario that may motivate other prior choices, given that the goal of choosing the model prior is not intrinsically related to the multiple-comparisons problem.

4 Cost-Adjusted Model Selection

Consider the standard, or “benefit-only”, variable selection problem introduced in Section 2.3. Regardless of the statistical or inferential paradigm, the goal is to identify the subset of predictors that best explains the data for estimation or prediction. However, many real-world applications involve additional considerations or constraints. In particular, in many settings, it is important to account for the characteristics associated with acquiring each explanatory variable in the candidate pool. These characteristics may reflect monetary costs, computational resources, modeling effort, data-collection time, or other practical constraints, and can be interpreted as costs associated with including a given predictor in the model. Accounting for this additional information implies that the selection criterion should account for these costs.

When multiple, potentially conflicting objectives are, there are two natural optimization approaches. The first is to consider all objectives simultaneously, explicitly modeling the trade-offs among them. The second is to maximize a single objective subject to constraints on the remaining ones. The first approach aligns naturally with the Bayesian model selection framework, which, as discussed previously, is intrinsically connected to decision theory. Within this framework, utility functions may define the selection criterion. Brown, Fearn, and Vannucci (1999), Fouskakis and Draper (2008), Lindley (1968), and Miyawaki and MacEachern (2023) modify traditional utility functions to incorporate “cost–benefit” considerations through the cost of including variables in the model. An example of the second approach is provided by Fouskakis, Ntzoufras, and Draper (2009a), who employed a more sophisticated MCMC algorithm to maximize predictive accuracy over the class of models satisfying a prespecified cost constraint.

This work focuses on medical applications. Clinical decision-making often takes place under resource constraints, where practical considerations play an important role. As a result, variable selection methods must account for the significant heterogeneous costs of collecting individual predictors while maintaining diagnostic accuracy, whether to satisfy budgetary constraints imposed by hospitals or insurance providers or simply reduce per-patient data-collection costs. Accordingly, the “gold standard” for diagnosing or explaining the response variable (e.g., a particular disease) may be impractical because the associated procedure is either prohibitively expensive or too invasive. In such situations, the goal is therefore to identify a subset of predictors that achieves comparable explanatory or predictive performance at a lower cost. In these scenarios, non-traditional approaches based on maximizing expected utility, such as that proposed by Fouskakis and Draper (2008), may be less suitable because the preferences underlying the specification of the utility function can differ across stakeholders (e.g., hospitals and insurers), making it difficult to define a common utility that adequately captures all relevant objectives. Moreover, it is desirable for the medical practitioner to control the degree of cost–performance trade-off, adjusting the level of cost penalization according to the available budget and the desired model performance. In practice, budgets may be flexible, performance-dependent or even undefined *a priori*, making this type of analysis particularly valuable. This motivates the need for an alternative

to the “cost-restrictive” approach of Fouskakis, Ntzoufras, and Draper (2009a). Although non-parametric cost-penalized methods have also been proposed and are capable of identifying lower-cost subsets of predictors, none (see Porter, Franck, and Adams, 2024, and references therein) appears to provide a single user-controlled tuning parameter that continuously adjusts the degree of cost penalization, as does the methodology described next.

4.1 Cost-Sensitive BIC

Originally introduced by Fouskakis, Ntzoufras, and Draper (2009b), cost-sensitive Bayesian Model Selection is based on the idea that the individual data-collection costs of explanatory variables can be incorporated into model selection through the prior distribution on the model space, naturally assigning different prior weights to models with different costs. As a result, models including more costly predictors receive greater penalization than they would under standard variable selection, where data-collection costs are not taken into account. Bayesian model selection then proceeds as in the traditional setting, using the posterior summaries HPM and MPM introduced previously for inference and prediction, with a preference for less costly subsets of predictors. Because the methodology trades off data-collection costs against predictive accuracy, it can be viewed as a principled approximation to the utility-based approach, without requiring its explicit specification.

This model prior (*FND*, following the terminology in Porter, Franck, and Adams, 2024) is constructed so that the resulting posterior model odds generalize the BIC (Schwarz, 1978). As a starting point, consider the case where the set of explanatory variables consists solely of numerical variables ($p = 0$). For any two models $\mathcal{M}_\gamma$ and $\mathcal{M}_{\gamma'}$ in $\mathcal{M}$, the posterior model odds of model $\mathcal{M}_{\gamma'}$ versus model $\mathcal{M}_\gamma$, $\text{PO}_{\mathcal{M}_{\gamma'}:\mathcal{M}_\gamma} = f(\gamma' | \mathbf{Y})/f(\gamma | \mathbf{Y})$, are the product of the corresponding Bayes factor and the prior odds, $f(\gamma')/f(\gamma)$. By straightforward manipulations, it follows that

$$-2 \log \text{PO}_{\mathcal{M}_{\gamma'}:\mathcal{M}_\gamma} = -2 \log \text{BF}_{\mathcal{M}_{\gamma'}:\mathcal{M}_\gamma} - 2 \log \frac{f(\gamma')}{f(\gamma)}. \quad (29)$$

From Section 3.2 and (29) the posterior odds can be viewed as a penalized log-likelihood ratio with two penalty terms: one arising from the Bayes factor, reflecting both Bayesian Ockham’s razor and the discrepancy between the data and the model parameters prior, and the another arising from the prior odds. Similiar in spirit to the Bayesian Ockham’s razor, the prior odds can provide a built-in penalty for costlier models by assigning cost-dependent penalties for individual predictors. The reasoning of Fouskakis, Ntzoufras, and Draper (2009b) was therefore to find a model prior such that, when costs are not considered, the prior odds are invariant across any pair of models, and when predictors’ data-collection costs differ considerably,

costlier models are penalized, *ceteris paribus*. Their proposal was:

$$f(\gamma) = \prod_{i=1}^k f(\gamma_i) = \prod_{i=1}^k \frac{\exp \left[-\frac{\gamma_i \log(n)}{2} \left(\frac{c_i}{c_0} - 1 \right) \right]}{1 + \exp \left[-\frac{\log(n)}{2} \left(\frac{c_i}{c_0} - 1 \right) \right]}, \quad (30)$$

where c_i denotes the individual costs of the predictors and $c_0 = \min\{c_i, i = 1, \dots, k\}$. Note that the prior inclusion probability of a given predictor, $f(\gamma_i = 1)$, is a decreasing function of c_i , being equal to $1/2$ when $c_i = c_0$. This formulation also makes the resulting decisions invariant to transformations of the cost scale (e.g., hours to minutes, euros to pounds, or hours to euros using a prevailing wage rate).

One may also consider an approximation of (29) that avoids the specification of priors for model parameters, is computationally simple, and provides reasonable results, although it may introduce bias into the posterior model odds, and consequently into the posterior model probabilities, even for large samples. Note that one can rewrite $f(\gamma \mid \mathbf{Y})$ in (6) as $\left(\sum_{\gamma' \in \mathcal{M}} \text{PO}_{\mathcal{M}_{\gamma'}, \mathcal{M}_{\gamma}} \right)^{-1}$. The approximation involves replacing the actual Bayes factor by the BIC for comparing models $\mathcal{M}_{\gamma'}$ and $\mathcal{M}_{\gamma}$, $\text{BIC}_{\mathcal{M}_{\gamma'}: \mathcal{M}_{\gamma}}$, a criterion that mediates the trade-off between model fit through the maximized likelihoods and model complexity through model sizes (Kass & Raftery, 1995):

$$\begin{aligned} -2 \log \text{PO}_{\mathcal{M}_{\gamma'}: \mathcal{M}_{\gamma}} &\approx -2 \log \left[\frac{f(\mathbf{Y} \mid \hat{\alpha}_{\gamma'}, \hat{\beta}_{\gamma'}, \mathcal{M}_{\gamma'})}{f(\mathbf{Y} \mid \hat{\alpha}_{\gamma}, \hat{\beta}_{\gamma}, \mathcal{M}_{\gamma})} \right] + (k_{\gamma'} - k_{\gamma}) \log(n) - 2 \log \frac{f(\gamma')}{f(\gamma)} \\ &\approx \text{BIC}_{\mathcal{M}_{\gamma'}: \mathcal{M}_{\gamma}} - 2 \log \frac{f(\gamma')}{f(\gamma)}. \end{aligned} \quad (31)$$

Then, using (30) in (31) leads to

$$-2 \log \text{PO}_{\mathcal{M}_{\gamma'}: \mathcal{M}_{\gamma}} \approx -2 \log \left[\frac{f(\mathbf{Y} \mid \hat{\alpha}_{\gamma'}, \hat{\beta}_{\gamma'}, \mathcal{M}_{\gamma'})}{f(\mathbf{Y} \mid \hat{\alpha}_{\gamma}, \hat{\beta}_{\gamma}, \mathcal{M}_{\gamma})} \right] + \frac{C_{\gamma'} - C_{\gamma}}{c_0} \log(n), \quad (32)$$

where $C_{\gamma} = \sum_{i=1}^k c_i \gamma_i$ is the total cost of an observation under model $\mathcal{M}_{\gamma}$. This means that, if all costs are equal, $C_{\gamma}/c_0 = k_{\gamma}$ and the resulting posterior odds in (32) have the same behavior as those induced by BIC. If predictor costs differ, the imposed penalty on the log-likelihood ratio can still be considered a complexity penalty, since the penalty for adding a variable is scaled by its relative cost c_i/c_0 relative to the BIC penalty $\log(n)$ that applies when $c_i = c_0$, *ceteris paribus*. For this reason, Fouskakis, Ntzoufras, and Draper (2009b) referred to (32) as a cost-adjusted generalization of BIC. This results in the cost-benefit analysis supporting more parsimonious models, in terms of cost, than the benefit-only analysis.

4.1.1 Flexible Penalization

Porter, Franck, and Adams (2024), however, highlighted a paradox that arises with the *FND* prior. On the one hand, one of the main advantages of traditional Bayesian methods is that the influence of the prior is often swamped as more data are collected. In this setting, however, this can undermine the original goal, as the induced posterior model probabilities can become similar to those obtained under the constant prior (for example, when measured in terms of the Kullback–Leibler divergence). This is somewhat intriguing because the prior probability of including a given variable $\mathbf{x}_i$ in (30), $f(\gamma_i = 1)$, is a decreasing function of the sample size, which converges to 0 as $n \rightarrow \infty$ (rather than to $1/2$). In some cases, however, the convergence is too slow to offset the increasing influence of the data on the model selection process.

The striking differences in the conclusions obtained with the *FND* prior across different sample sizes, and hence across different datasets, led Porter, Franck, and Adams (2024) to propose replacing the relative costs c_i/c_o in (30) with functions of these ratios (e.g., $(c_i/c_o)^b$, which is the specification adopted throughout this work), while preserving cost-conversion invariance. By introducing the user-controlled tuning parameter b , the methodology gains additional flexibility that is useful not only in situations such as the one described above. In particular, choosing values of $b > 1$ increases the prior cost penalization relative to the *FND* prior ($b = 1$), thereby preventing practitioners from ordering more expensive procedures than necessary or exceeding the available budget. Conversely, when the model selected under the *FND* prior performs below acceptable standards, choosing $0 < b < 1$ allows the selection of an alternative set of predictors with improved predictive performance while still penalizing costlier models, unlike the “benefit-only” approach ($b = 0$). Naturally, when $b \neq 1$, the posterior model odds are no longer BIC-like, although they retain a similar structure and the resulting posterior model probabilities continue to provide a cost-sensitive assessment of competing models.

More generally, this flexible approach enables cost-adjusted analyses over a range of values of b , making it possible to tailor the degree of cost penalization to the dataset and clinical setting. Consequently, medical researchers and practitioners can explicitly control the trade-off between predictive performance and data-collection burden while retaining the interpretability of posterior probabilities for both models and individual predictors.

4.1.2 Categorical Variables

The prior in (30) reduces to the prior assigning equal probabilities to all candidate models when all predictors have the same data-collection cost. Although this prior is motivated by inducing BIC-like behaviour in the posterior model probabilities, it does not account for the multiple-testing problem discussed in Section 2.4. It is not the purpose of this work to propose an alternative to (30) that addresses multiplicity through a prior such as (11), since the underlying assumptions of the two priors are fundamentally different. First, the prior in (11) depends only on model size and the number of competing models of that size, and it is not clear how it could be adapted

to incorporate predictor-specific external information such as data-collection costs (recall that (11) is obtained by placing a *hyperprior* on a common inclusion probability, w). Likewise, constructing an analogous prior inversely proportional to the number of models having the same total cost would neither be straightforward nor particularly well motivated. More fundamentally, any attempt to control for multiplicity in this setting would necessarily depend on the cost structure of the candidate predictors. Accordingly, rather than proposing a new multiplicity-adjusted prior, this work highlights this limitation and encourages practitioners to take it into account, particularly when the candidate set contains baseline-cost predictors with uncertain effects, whose prior inclusion probability remains $1/2$. In contrast, spurious predictors with collection costs above the baseline are naturally penalized by the prior, with this penalization increasing as b increases.

In the experiments conducted by Fouskakis, Ntzoufras, and Draper (2009b) and Porter, Franck, and Adams (2024), the corresponding datasets included only numerical variables and, in some cases, dichotomous variables (factors with $\ell_j = 2$ levels) or ordinal variables (with three or more categories) measured on quantitative scales. In all cases, each variable entered the model through a single coefficient. For ordinal variables, the assignment of numerical scores to the categories, as in these examples, is often based on subjective, and sometimes arbitrary, considerations. Moreover, this treatment implicitly assumes a linear effect on the linear predictor, so that the relevance being assessed is that of proportional differences between the effects of consecutive categories, an assumption that may not hold in practice. Representing such variables instead as categorical factors removes this restriction, allowing each category to have its own effect and thereby providing greater modeling flexibility. Although this approach increases the number of parameters and may yield estimated category effects that do not respect the natural ordering of the levels, it avoids imposing potentially inappropriate “linearity” assumptions. In addition, examining the estimated category effects may provide guidance on a suitable numerical scoring for the categories. These considerations also motivate extending the methodology introduced in Section (3.1) to Cost-Adjusted Bayesian Model Selection to accommodate categorical variables with $\ell_j \geq 2$.

One of the assumptions of the framework introduced by Fouskakis, Ntzoufras, and Draper (2009b) is that the data-collection cost is associated with the predictor itself, rather than with the value it takes. This assumption is not particularly restrictive in practice. For example, a more sophisticated measuring instrument may provide greater precision or may be required to reliably detect observations in certain regions of the predictor’s range, such as extreme values. Similarly, for categorical variables, a larger number of levels may be recorded to achieve greater granularity. In either case, the practitioner can assign to the predictor the cost of the measurement procedure. Thus, the cost remains associated with the predictor rather than with its realized value. This observation naturally extends the cost penalization to categorical predictors by assigning a single cost to the variable as a whole, rather than to each of its associated dummy variables. Accordingly, the proposed methodology can be combined with the approach of García-Donato and

Paulo (2022b) by replacing (30) with the corresponding “marginal prior”, yielding

$$f(\boldsymbol{\gamma}, \boldsymbol{\tau}) \propto \prod_{i=1}^k \exp \left[-\frac{\gamma_i \log(n)}{2} \left(\frac{c_i}{c_0} - 1 \right) \right] \prod_{j=1}^p \exp \left[-\frac{\tau_j \log(n)}{2} \left(\frac{c_j}{c_0} - 1 \right) \right], \quad (33)$$

where the baseline cost is now defined as $c_0 = \min \left(\{c_i\}_{i=1}^k \cup \{c_p\}_{j=1}^p \right)$. Under the two-stage prior framework, (31) becomes

$$\begin{aligned} -2 \log \text{PO}_{\mathcal{M}_{\boldsymbol{\gamma}', \boldsymbol{\delta}'} : \mathcal{M}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}} &= -2 \log \left[\frac{f(\mathbf{Y} \mid \hat{\alpha}_{\boldsymbol{\gamma}', \boldsymbol{\delta}'}, \hat{\beta}_{\boldsymbol{\gamma}', \boldsymbol{\delta}'}, \mathcal{M}_{\boldsymbol{\gamma}', \boldsymbol{\delta}'})}{f(\mathbf{Y} \mid \hat{\alpha}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}, \hat{\beta}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}, \mathcal{M}_{\boldsymbol{\gamma}, \boldsymbol{\delta}})} \right] \\ &\quad + (k_{\boldsymbol{\gamma}', \boldsymbol{\delta}'} - k_{\boldsymbol{\gamma}, \boldsymbol{\delta}}) \log(n) - 2 \log \frac{f(\boldsymbol{\gamma}', \boldsymbol{\delta}')}{f(\boldsymbol{\gamma}, \boldsymbol{\delta})} + O(1) \\ &= \text{BIC}_{\mathcal{M}_{\boldsymbol{\gamma}', \boldsymbol{\delta}'} : \mathcal{M}_{\boldsymbol{\gamma}, \boldsymbol{\delta}}} - 2 \log \frac{f(\boldsymbol{\gamma}', \boldsymbol{\tau}')}{f(\boldsymbol{\gamma}, \boldsymbol{\tau})} - 2 \log \frac{f(\boldsymbol{\delta}' \mid \boldsymbol{\gamma}', \boldsymbol{\tau}')}{f(\boldsymbol{\delta} \mid \boldsymbol{\gamma}, \boldsymbol{\tau})} + O(1). \end{aligned} \quad (34)$$

As expected, when $c_i = c_0$ for all i and $c_j = c_0$ for all j , (33) reduces to the marginal constant prior considered by García-Donato and Paulo (2022b), $f(\boldsymbol{\gamma}, \boldsymbol{\tau}) = 2^{-(k+p)}$. Note that, under the two-stage framework, the cost-adjusted prior inclusion probabilities are still decreasing functions of the corresponding predictor costs, with baseline-cost predictors retaining prior inclusion probability 0.5 through (33). Moreover, these prior inclusion probabilities are independent of the number of levels of a categorical predictor, unlike if (30) were extended directly to $f(\boldsymbol{\gamma}, \boldsymbol{\delta})$. To illustrate, consider a categorical predictor with $\ell_2 = 3$ levels and data-collection cost equal to the baseline. Table S.4 of García-Donato and Paulo (2022a) shows that, although every component of $\boldsymbol{\delta}_2$ equals 1 in fewer than half of the models in $\mathcal{M}$, $\tau_2 = 1$ in more than half of them. This discrepancy becomes more pronounced as the number of levels increases.

When there were no factors, equal costs among predictors translated into equal prior odds for any pair of models in $\mathcal{M}$, and induced BIC-like behavior in the posterior odds, through (31). With the introduction of factors, equal predictor costs still imply $f(\boldsymbol{\gamma}', \boldsymbol{\tau}')/f(\boldsymbol{\gamma}, \boldsymbol{\tau}) = 1$. However, the posterior odds now include the additional term $f(\boldsymbol{\delta}' \mid \boldsymbol{\gamma}', \boldsymbol{\tau}')/f(\boldsymbol{\delta} \mid \boldsymbol{\gamma}, \boldsymbol{\tau})$, which may differ from one under either of the conditional prior specifications studied by García-Donato and Paulo (2022b). For the SB-type conditional prior, take a simple problem with two factors only ($p = 2$ and $k = 0$) with $\ell_1 = \ell_2 = 3$. Let $\mathcal{M}_{\boldsymbol{\delta}'}$ and $\mathcal{M}_{\boldsymbol{\delta}}$ be two competing models in this model set $\mathcal{M}$ such that $\boldsymbol{\tau}' = \boldsymbol{\tau} = (1, 1)$ (both factors are active), differing only in the number of levels of the categorical variables ($\boldsymbol{\delta}' \neq \boldsymbol{\delta}$). Then, the ratio of the conditional prior probabilities may be greater than, equal to, or less than one, depending on the model rank of the competing models. In essence, this means that the prior odds, sometimes favor models of higher rank, something that did not happen under the “benefit-only” approach with numerical variables only.

Turning to the Constant-type conditional prior ($f(\boldsymbol{\delta} \mid \boldsymbol{\gamma}, \boldsymbol{\tau}) = |\mathcal{M}(\boldsymbol{\gamma}, \boldsymbol{\tau})|^{-1}$; see García-Donato and Paulo, 2022b), it also leads to some curious conclusions. From

Result S.2 of Section S.4 of García-Donato and Paulo (2022a), $\sum_{j=1}^p \tau_j$ increases the size of $\mathcal{M}(\gamma, \tau)$, if and only if, the added active factors satisfy $\ell_j \geq 3$. In particular, this means that among the models that share the same combination of active factors ($\tau' = \tau$), there is no complexity penalty distinguishing them coming through the prior probabilities, similar to BIC-like posterior model odds. However, consider situations where $\sum_{j=1}^p (\tau'_j - \tau_j) \geq 1$ and $k_{\gamma', \delta'} > k_{\gamma, \delta}$, with at least one of the extra active factors satisfying $\ell_j \geq 3$. Given that $|\mathcal{M}(\gamma', \tau')| > |\mathcal{M}(\gamma, \tau)|$, the overall complexity penalty is higher than the one induced by BIC. In contrast, when $k_{\gamma', \delta'} < k_{\gamma, \delta}$, the opposite follows, i.e., the overall complexity penalty is lower than the one implied by BIC.

In order to better understand the implications of both conditional priors for the cost-adjusted selection process, they are studied in the practical experiment of this chapter. (33) can also be generalized to accommodate functions of the cost ratios, such as $(c_j/c_0)^b$.

4.1.3 Cost Structures

Both the works of Fouskakis, Ntzoufras, and Draper (2009b) and Porter, Franck, and Adams (2024) and the proposed framework have so far assumed that each predictor has an individual data-collection cost that is independent of the costs of all other predictors. In practice, this does not always reflect the way data are collected. In particular, the cost of acquiring a predictor often depends on whether other predictors are collected simultaneously. This is especially common in clinical settings, where a single data-collection procedure may provide several predictors. For example, a single blood sample can be used to measure multiple biomarkers, so that once the sample has been collected, the additional cost of obtaining further measurements is negligible. Likewise, a diagnostic procedure such as an exercise electrocardiogram may produce several clinically relevant variables, with the principal cost arising from performing and interpreting the examination rather than from recording each individual measurement. In these situations, assigning an independent cost to every predictor overestimates the true data-collection burden. More generally, predictors may exhibit structured cost dependencies, in the sense that they become less expensive to collect when considered jointly owing to shared equipment, personnel, or preparation costs. Below, the proposed cost-adjusted model prior is extended to accommodate these situations more effectively. This extension enables the framework to account for shared collection costs while preserving the cost-sensitive nature of Bayesian model selection.

For ease of exposition, (30) is extended, although the resulting construction generalizes directly to the marginal model prior in (33). In the absence of cost structures, the model prior can be modeled through the product of cost-penalized priors for single inclusion indicators, $f(\gamma_i)$. In the presence of cost structures, this assumption no longer holds. Again, for ease of exposition, it is assumed that the cost structures arise only for a single group of predictors, $\mathbf{A} = \{\mathbf{x}_1, \dots, \mathbf{x}_{|A|}\}$, with

$2 \leq |A| \leq k$. This gives

$$f(\gamma) = f_{\mathbf{A}}(\gamma_1, \dots, \gamma_{|\mathbf{A}|}) \prod_{i=|\mathbf{A}|+1}^k f(\gamma_i), \quad (35)$$

where $f(\gamma_i)$ follows from (30).

Throughout this development, only two-stage costs dependencies are considered, although more complex cost structures may arise in practice. So far, c_i has been defined as the marginal data-collection cost per observation for predictor $\mathbf{x}_i$. For $i = 1, \dots, |\mathbf{A}|$ it is now interpreted as the full cost, incurred when the corresponding predictor is the “first” test in a group of procedures to be ordered for a given patient. For these *discounted* predictors, c_i^* denotes the discounted cost, charged when the corresponding predictor is the “second” or a “subsequent” within the same group to be ordered, with $c_0 < c_i^* < c_i$. For predictors obtained from a common procedure, such as blood sampling, termed *grouped* predictors, c_i represents the cost of that procedure and c_i^* is set equal to c_0 to reflect that no additional cost is incurred by including the predictor in the model, provided that another predictor from the same procedure has already been selected. Furthermore, it is assumed that the baseline cost, c_0 , remains unchanged after introducing cost structures into the analysis. In addition, it is assumed that $c_i + \sum_{s \in \mathbf{A} \setminus \{i\}} c_s^*$ takes the same value for all i in $\mathbf{A}$. Indeed, if this quantity were smaller for some predictor i , the discounted costs c_s^* of the remaining predictors in $\mathbf{A}$ could be redefined so that the equality holds. This follows because the framework considers an offline decision-making setting, in which model selection takes place before any tests are ordered and all collection costs are assumed to be known in advance. Consequently, only the minimum total cost of collecting a given set of predictors is relevant, irrespective of the order in which the corresponding tests are performed. Based on these considerations, $f_{\mathbf{A}}(\gamma_1, \dots, \gamma_{|\mathbf{A}|})$ in (35) is defined as

$$f_{\mathbf{A}}(\gamma_1, \dots, \gamma_{|\mathbf{A}|}) \propto \max_{i \in \mathbf{A}} \left\{ f^*(\gamma_i) \cdot \prod_{\substack{s \in \mathbf{A} \\ s \neq i}} f_d^*(\gamma_s \mid \gamma_i) \right\}, \quad (36)$$

where

$$\begin{aligned} f^*(\gamma_i) &= \exp \left\{ -\frac{\gamma_i \log(n)}{2} \left(\left(\frac{c_i}{c_0} \right)^b - 1 \right) \right\}, \\ f_d^*(\gamma_s \mid \gamma_i) &= \exp \left\{ -\frac{\gamma_s \log(n)}{2} \left(\left(\frac{c_s(1 - \gamma_i) + c_s^* \gamma_i}{c_0} \right)^b - 1 \right) \right\}. \end{aligned} \quad (37)$$

Note that neither $f^*(\gamma_i)$ nor $f_d^*(\gamma_s \mid \gamma_i)$ is itself a probability distribution. Interestingly, when $b = 1$, $f^*(\gamma_i)$ coincides with the numerator of $f(\gamma_i)$ in (30). It is also straightforward to verify that, in the absence of cost dependencies among the predictors in $\mathbf{A}$ (i.e., when $c_s = c_s^*$ for all s in $\mathbf{A}$), $f_d^*(\gamma_s \mid \gamma_i)$ reduces to $f^*(\gamma_i)$, and (36) simplifies to the cost-adjusted prior in (30). Equation (36) defines a probability

distribution up to a normalizing constant. However, this constant does not need to be evaluated, since it cancels in the computations of the prior odds in (17) and, consequently, in (29). The resulting prior therefore induces a cost-sensitive model comparison criterion that generalizes BIC when $b = 1$. Theoretical properties of the proposed prior, together with a simulation study, are presented in the supplementary material. The methodology is also illustrated in the case study of the next section.

4.2 Case Study

The experiment presented in this section evaluates the proposed approaches for incorporating categorical variables and cost structures into cost-penalized Bayesian model selection. The setting is a logistic regression (P. K. Dunn & Smyth, 2018), performed on a real dataset of clinical test results from the Cleveland Clinic Foundation, where the target variable indicates whether a patient has heart disease. More details on this dataset and the corresponding candidate predictors are provided in Section 3.4 of Porter, Franck, and Adams (2024). Information regarding the costs of the predictors (in Canadian dollars) and the corresponding cost structures is presented in Table 2.

Figures 2 and 3 display the results using the inclusion path technique proposed by Porter, Franck, and Adams (2024), which provides a convenient visualization to compare cost-adjusted analyses and select cost-effective predictors for heart disease diagnosis under different levels of the tuning parameter b . Twenty-one cost-adjusted analyses were performed for each proposal, with b ranging from 0 to 0.6 in increments of 0.03. No changes in the results were observed for values of b above 0.6. The settings of the Gibbs sampler are the same as those used in Section 3.3.2. In addition to the cost-adjusted PIPs, the costs of the selected models and their corresponding in-sample predictive performance are reported. Predictive performance is measured by the Area Under the Receiver Operating Characteristic (ROC) curve, also known as the Concordance (C) statistic (Porter, Franck, & Adams, 2024).

The prior adopted for the model parameters in this experiment was the LIM g-prior in (24) with $g = n$, reflecting its close relationship with the BIC. Large-sample asymptotic arguments in Li and Clyde (2018) indicate that the approximate conditional posterior distributions of the model parameters converge to a multivariate normal distribution. Consequently, the posterior mode is a function of the MLE of the coefficients, adjusted by the shrinkage factor $g/(1 + g)$ for $\hat{\beta}_{\gamma, \delta}$. This result is particularly useful because it enables efficient computation of the fitted values under both the HPM and the MPM, facilitating the evaluation of their corresponding C-statistics.

As expected, the results differed only marginally between the two conditional priors adopted for factor inclusion, as the competing factors have at most $\ell = 4$ levels. Only subtle differences were observed. Interestingly, the predictive performance was comparable to that reported by Porter, Franck, and Adams (2024), despite the substantially different treatment of categorical variables. This suggests that the categorization adopted for the ordinal predictors was appropriate, likely reflecting

Variable	Marginal Cost	Discounted Cost	Group
age	1.00	–	–
sex	1.00	–	–
chest pain	1.00	–	–
resting BP	1.00	–	–
cholesterol	7.27	5.17	A
blood sugar	5.20	3.10	A
EKG	15.50	–	–
heart rate	102.90	1.00	B
exercise angina	87.30	1.00	C
ST depression	87.30	1.00	C
peak ST segment	87.30	1.00	C
major vessels	100.90	–	–
defect type	102.90	1.00	B

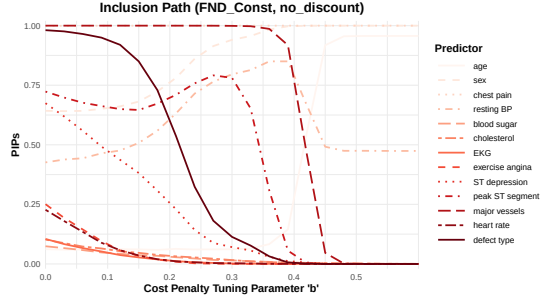
Table 2: Marginal and discounted acquisition costs by predictor. Variables with no group discount are marked with –.

expert knowledge. Regarding the introduction of cost structures, both approaches naturally produced identical results in the benefit-only analysis ($b = 0$). For larger values of b , however, the differences become more interesting. Contrary to our initial expectations, the discounted approach did not consistently outperform the standard cost specification in terms of predictive accuracy across all values of b . With respect to model cost, some of the observed differences arise from genuine differences in the selected models, whereas others simply reflect the different definitions used to compute predictor costs under each cost structure.

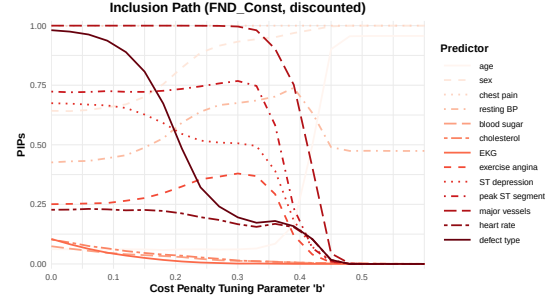
The introduction of factors into the set of candidate variables also requires re-defining the notion of the MPM. One possibility is to define the MPM as the model containing all numerical predictors with PIPs greater than 0.5, together with only those dummy variables corresponding to factor levels whose PIPs also exceed 0.5. This definition is referred to as the MPM dummies in the figures. Alternatively, because the prior inclusion probability is assigned at the factor level and fixed at 0.5, the corresponding prior inclusion probabilities of the individual dummy variables are implicitly penalized, with the penalty increasing as the number of factor levels grows. This motivates an alternative definition, termed the MPM grouped, in which all levels of a categorical variable are included whenever the factor itself has a posterior inclusion probability greater than 0.5. By construction, the MPM grouped is always at least as large as the MPM dummies. The two models may differ substantially when some levels of a factor have PIPs below 0.5 while the overall pos-

terior inclusion probability of the factor exceeds 0.5. In cost-penalized settings, such differences may also translate into different model costs. Furthermore, even when both models have identical costs, comparing them remains informative because their relative usefulness depends on the application (see, for example, the distinct settings considered in Experiments 1 and 2 of Chapter 3). Moreover, discrepancies between the two models may indicate that certain factor levels contribute little to prediction, providing evidence for potentially merging levels, as discussed by García-Donato and Paulo (2022b). The results shown in Figures 2 and 3 indicate that, whenever the two definitions differ, the grouped MPM generally achieves higher predictive accuracy, albeit at the expense of selecting more expensive models.

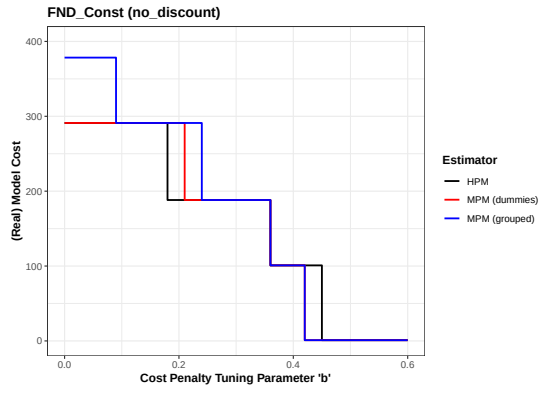
Overall, the proposed methodology is capable of identifying models that achieve substantial reductions in acquisition cost while incurring only minor losses in predictive performance for appropriate values of the tuning parameter b .



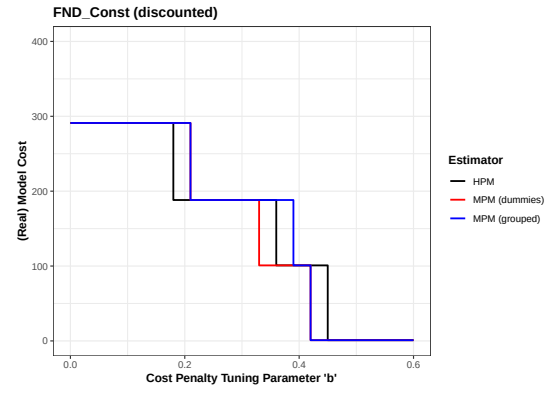
(a)



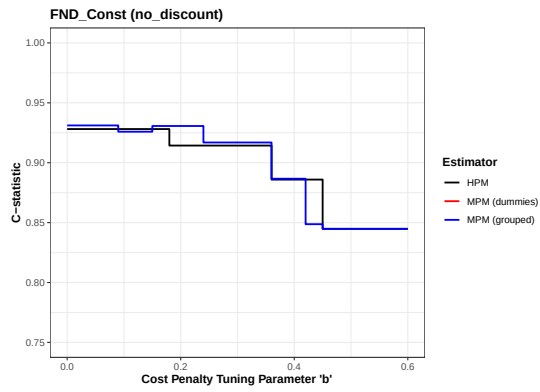
(b)



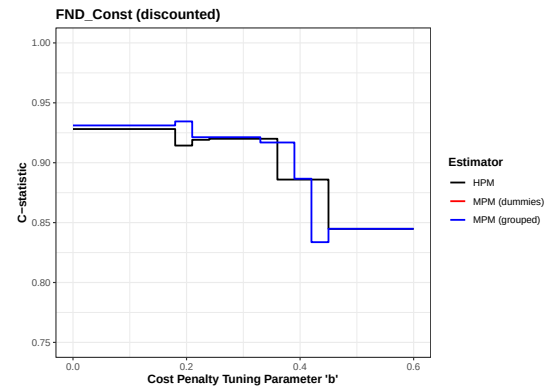
(c)



(d)



(e)



(f)

Figure 2: Inclusion paths (a, b), Model Cost (c, d), and C-statistic (e, f) under the Constant conditional prior. **Left:** no-discount analysis; **right:** discounted analysis.

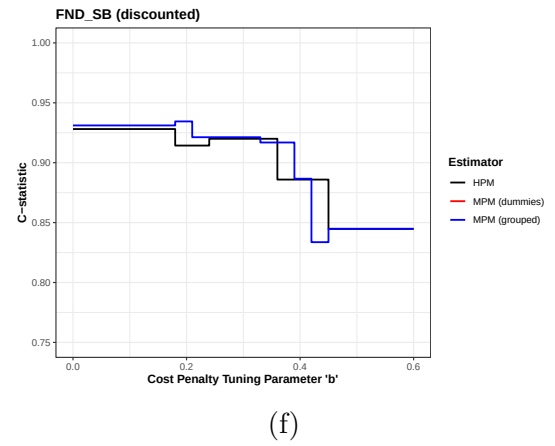
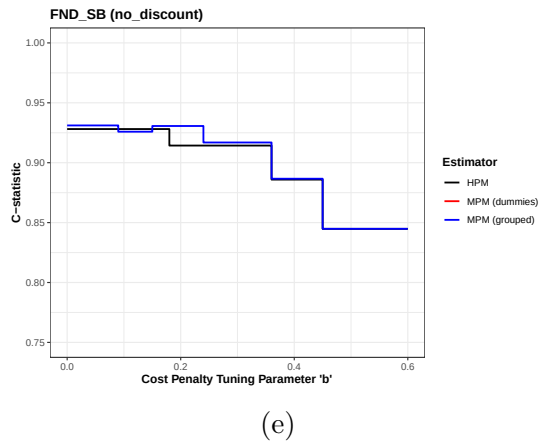
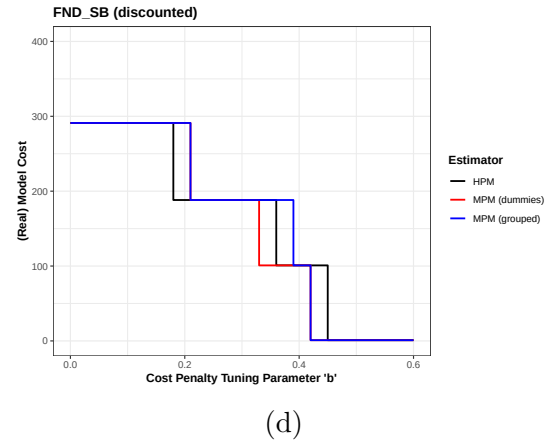
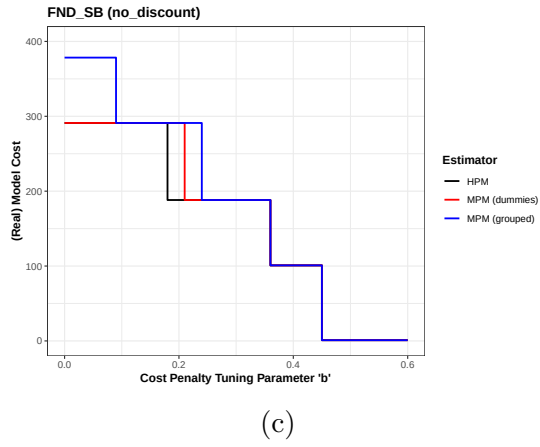
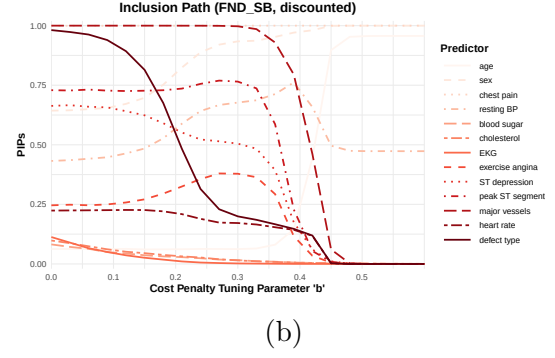
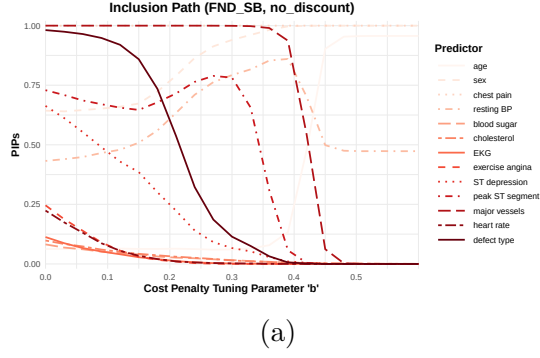


Figure 3: Inclusion paths (a, b), Model Cost (c, d), and C-statistic (e, f) under the SB-type conditional prior. **Left:** no-discount analysis; **right:** discounted analysis.

5 Conclusions

This dissertation extended cost-adjusted Bayesian variable selection to settings in which the pool of candidate predictors contains factors and in which data-collection costs are not mutually independent. Two results underpin the construction. First, the approximate null-based Bayes factors obtained through iL approximations under the LIM and CHIC g-priors were shown to be invariant to the choice of reference levels, which justifies carrying the methodology of García-Donato and Paulo (2022b) over to GLMs. Second, the experiments of Chapter 3 confirmed that SB-type marginal priors retain their multiplicity-control properties in log-linear model. The evidence separating the two conditional priors, by contrast, proved too volatile to be conclusive. Chapter 4 then attached the cost penalty to a factor as a whole, keeping prior inclusion probabilities free of its number of levels, and accommodated grouped and discounted predictors through a prior that reduces to that of Fouskakis, Ntzoufras, and Draper (2009b) in the absence of cost dependencies. Applied to the Cleveland data, the procedure identified models achieving substantial reductions in acquisition cost at only minor loss in discriminatory performance, although the discounted specification did not uniformly dominate its independent-cost counterpart.

5.1 Difficulties in Designing Simulation Studies

Regarding the simulation experiments of Chapter 3, binomial and exponential EDMs were also tested, but the results were not favorable in terms of identifying the active factors for $n = 300$. Two sources of difficulty may be at play. The first concerns Laplace’s approximation. The error of the Laplace approximation used in the computation of the Bayes factors is $O(n^{-1})$, but this order says nothing about which EDM likelihood is being approximated: the quality of the approximation at fixed n varies across families (Slate, 1994), reflecting differences in the geometry of the underlying EDM. The second lies in the design of the experiments. The interpretation of the model parameters depends on the link function adopted in (3), and the values assigned to the coefficients should be thought of in terms of the unit deviance (see Ch. 5 of P. K. Dunn and Smyth (2018)). What governs detectability is not the separation between observations on the scale of the response, but their separation as measured by the unit deviance. For the normal EDM this reduces to $(y - \mu)^2$, the familiar Euclidean distance; outside the normal case, it is asymmetric and does not follow any standard form. Consequently, since total deviances are not comparable across families, it is not straightforward to calibrate simulation scenarios using the linear regression case as a benchmark. Moreover, even the choice of the intercept affects the results, unlike in linear regression. This is because the variance function of the normal EDM does not depend on the mean, whereas this is not the case for other EDMs. Consequently, changing the intercept alters not only the mean response but also the variance structure, thereby affecting the detectability of active factors.

5.2 Computation and Supplement Material

The supplementary material and the R and C code used for the simulations and analysis of the real dataset can be found at <https://github.com/davidembb20>. The functions supporting Bayesian Model Selection are available in a branch of the R package *BAS* (M. Clyde, 2025), hosted on GitHub. Some of the implementations in this branch were based on the R package *BayesVarSel* (Gonzalo Garcia-Donato & Anabel Forte, 2018), used for the experiments in García-Donato and Paulo (2022b). Regarding the implementation of the Gibbs Sampler, hash tables (tree-based data structures) were used to store the visited models (including intermediate models), thereby avoiding repeated evaluations of the marginal likelihood. All computations were performed on a server provided by ISEG Research, running Fedora Linux 44 (kernel 7.1.3), equipped with an AMD Ryzen Threadripper 3960X processor (24 cores / 48 threads) and 64 GB of RAM. Although parallel computing was used to improve computational performance, its benefits were limited, given that MCMC algorithms are inherently sequential and therefore cannot be efficiently parallelized. In contrast, parallelization increased memory consumption, and in some cases the server’s RAM approached its maximum capacity, requiring the number of worker processes to be reduced.

5.3 Future Research

Regarding the implementation of the Gibbs Sampler, the presence of factors introduces additional constraints. Note that the draws from the full conditional distributions in (16) do not account for whether the current model or the proposed model for the next step belongs to the real model space $\mathcal{M}$, defined in Section 3.1. Consequently, the model space explored by the Gibbs Sampler is the one obtained from the overparameterized design matrix, including repeated models. As a result, the prior model probabilities used for the Gibbs sampler are not those obtained by applying the expressions in García-Donato and Paulo (2022b), but instead an adjusted version. To account for this discrepancy, a weighted resampling step was performed after the Gibbs sampler had finished. The resampling weights were chosen to be proportional to the true prior model probabilities, while assigning zero weight to models outside the real model space $\mathcal{M}$. Consequently, only valid models were retained in the final set of visited models. Nonetheless, it remains unclear whether this procedure introduces bias into the estimator of the posterior model probabilities or whether the search mechanism can be modified to explore only the valid model space while preserving the Gibbs sampler’s theoretical properties.

A natural avenue for future research is the extension of the proposed methodology to sequential decision-making, a direction identified by Porter, Franck, and Adams (2024) that appears to remain largely unexplored. The motivating setting is medical diagnosis, where practitioners order additional tests depending on a patient’s initial results rather than committing to a full battery of measurements at the outset. Ji and Carin (2007) formalize this as a process in which inexpensive tests are collected first, disease probabilities are updated, and more expensive tests are ordered only

while the information they provide justifies their cost. Their reasoning offers a natural starting point for developing such an extension within the Bayesian model selection framework, which would bring the analysis closer to the way decisions are actually made in clinical practice.

Regarding the cost structures introduced in Section 4.1.3, variables arising from the same procedure (grouped predictors) may carry closely related information. This suggests that the prior inclusion probability of such a group could itself be controlled, in the spirit of the dilution priors of H. A. Chipman (1996), a possibility not accounted for here.

A Appendix

Proof: Bayes Factor Reference-Level Invariance

Here it is shown that, for an overparameterized model $\mathcal{M}_{\gamma,\delta}$ in $\mathcal{M}$, the set of unique models obtained through the procedure described in Section 3.1, the approximate null-based Bayes factors in (26) and in Section 3.1 of Li and Clyde (2018), computed for the duplicate models of $\mathcal{M}_{\gamma,\delta}$ that are excluded from $\mathcal{M}$, are invariant to a change of reference levels. These approximate Bayes factors vary across the models in $\mathcal{M}$, as they depend on model-specific quantities: the number of columns of the model-specific design matrix, $k_{\gamma,\delta}$; the observed Fisher information for the intercept, $\mathcal{J}_n(\hat{\alpha}_{\gamma,\delta})$; and the deviance and Wald statistics, $z_{\gamma,\delta}$ and $Q_{\gamma,\delta}$, respectively.

Under treatment coding, a full-rank parameterization of (22) for $\mathcal{M}_{\gamma,\delta}$ is

$$\mathcal{M}_{\gamma,\delta'} : \boldsymbol{\eta}_{\gamma,\delta'} = \mathbf{1}_n \alpha + \mathbf{X}_\gamma \boldsymbol{\beta}_\gamma + \mathbf{Z}_{\delta'}^* \boldsymbol{\lambda}_{\delta'} = \mathbf{1}_n \alpha + \tilde{\mathbf{X}}_{\gamma,\delta'} \boldsymbol{\beta}_{\gamma,\delta'}, \quad (38)$$

with $\boldsymbol{\beta}_{\gamma,\delta'} = (\boldsymbol{\beta}_\gamma, \boldsymbol{\lambda}_{\delta'})^\top$ and $\tilde{\mathbf{X}}_{\gamma,\delta'} = [\mathbf{X}_\gamma \mid \mathbf{Z}_{\delta'}^*]$, where $\mathbf{Z}_{\delta'}^*$ is obtained from $\mathbf{Z}_\delta$ in (22) by removing the columns corresponding to the chosen reference levels, and δ' is obtained from δ by setting the entries corresponding to those reference levels equal to 0. Clearly, $\mathcal{M}_{\gamma,\delta'} \notin \mathcal{M}$. Also, let $p_{\delta'}$ denote the number of columns of the full-column-rank matrix $\mathbf{Z}_{\delta'}^*$, i.e., $p_{\delta'} = \sum_{j=1}^p \sum_{r=1}^{\ell_j} \delta'_{jr}$.

For an invertible $(1 + k_\gamma + p_{\delta'}) \times (1 + k_\gamma + p_{\delta'})$ matrix $\mathbf{R}$, a change of reference levels corresponds to a reparameterization of (38) given by

$$\begin{aligned} \mathcal{M}_{\gamma,\delta''} : \boldsymbol{\eta}_{\gamma,\delta''} &= ([\mathbf{1}_n \mid \tilde{\mathbf{X}}_{\gamma,\delta'}] \cdot \mathbf{R}) (\mathbf{R}^{-1} \cdot (\alpha, \boldsymbol{\beta}_{\gamma,\delta'})^\top) \\ &= \mathbf{1}_n \alpha^{\mathbf{R}} + \tilde{\mathbf{X}}_{\gamma,\delta''} \boldsymbol{\beta}_{\gamma,\delta''} = \boldsymbol{\eta}_{\gamma,\delta''}, \end{aligned} \quad (39)$$

with $\boldsymbol{\beta}_{\gamma,\delta''} = (\boldsymbol{\beta}_\gamma, \boldsymbol{\lambda}_{\delta''})^\top$ and $\tilde{\mathbf{X}}_{\gamma,\delta''} = [\mathbf{X}_\gamma \mid \mathbf{Z}_{\delta''}^*]$. The matrix $\mathbf{R}$ is chosen to be of the form:

$$\mathbf{R} = \begin{bmatrix} 1 & \mathbf{d}^\top \\ 0 & \mathbf{D} \end{bmatrix}, \quad \mathbf{R}^{-1} = \begin{bmatrix} 1 & -\mathbf{d}^\top \mathbf{D}^{-1} \\ 0 & \mathbf{D}^{-1} \end{bmatrix}, \quad \mathbf{D} = \begin{bmatrix} \mathbf{I}_{k_\gamma} & 0 \\ 0 & \mathbf{A} \end{bmatrix}, \quad \mathbf{d}^\top = (0_{k_\gamma}^\top, \mathbf{c}^\top),$$

where $\mathbf{c}$ is a $p_{\delta'}$ -dimensional vector whose entries equal 1 for the columns corresponding to the former reference levels and 0 elsewhere, reflecting that the effect of each former reference level is absorbed into the intercept after reparameterization. This relates the design matrices of the two full-rank reparameterizations via $\tilde{\mathbf{X}}_{\gamma,\delta''} = \tilde{\mathbf{X}}_{\gamma,\delta'} \mathbf{D} + \mathbf{1}_n \mathbf{d}^\top$ and $\boldsymbol{\beta}_{\gamma,\delta''} = \mathbf{D}^{-1} \boldsymbol{\beta}_{\gamma,\delta'}$. Note also that $\mathbf{Z}_{\delta''}^* = \mathbf{Z}_{\delta'}^* \mathbf{A} + \mathbf{1}_n \mathbf{c}^\top$. It follows that δ'' is obtained from δ' by setting the old reference levels to zero and the new ones to 1, and that $p_{\delta''} = \text{rank}(\mathbf{Z}_{\delta'}^*) = \text{rank}(\mathbf{Z}_{\delta''}^*) = p_{\delta'}$. The linear transformation $\mathbf{R}$ differs from those used to establish measurement invariance of the LIM and CHIC g-priors (see Section 4.2 of Li and Clyde, 2018). It likewise differs from the scenario in Section 2.5 of Li and Clyde (2018). Although $\mathcal{M}_{\gamma,\delta}$ is a non-identifiable model, its design matrix $[\mathbf{X}_\gamma \mid \mathbf{Z}_\delta]$ is assumed to have full-column rank. Neverthe-

less, the arguments used here are similar, as are the conclusions, yielding a further invariance property.

Note that a change of reference levels does not alter the linear predictor $\boldsymbol{\eta}_{\gamma, \delta'}$. Unlike in linear regression, however, the score equations used for maximum likelihood estimation of model parameters for other GLMs most often do not have an explicit solution. In such cases, the MLEs are computed numerically, using the *Fisher scoring* algorithm (P. K. Dunn & Smyth, 2018). Since the goal is to produce estimates close to the observed values, y_i , the algorithm is initialized close to y_i , and this initialization does not depend on the chosen parameterization. For this reason, the estimated values of the linear predictor at each iteration are parameterization-invariant. Additionally, the working coefficients (see Ch. 6 of P. K. Dunn and Smyth, 2018) satisfy the relation $\boldsymbol{\beta}_{\gamma, \delta''} = \mathbf{D}^{-1} \boldsymbol{\beta}_{\gamma, \delta'}$ at each iteration, including the last, leading to $\hat{\boldsymbol{\beta}}_{\gamma, \delta''} = \mathbf{D}^{-1} \hat{\boldsymbol{\beta}}_{\gamma, \delta'}$. Consequently, the MLE $\hat{\boldsymbol{\eta}}_{\gamma, \delta'}$ remains unchanged under the reparameterization defined in (39), so that, from (28), it follows immediately that $z_{\gamma, \delta}$ is invariant under this type of reparameterization. Results (10), (11) and (13) of Li and Clyde (2018) lead to the same conclusion for $\mathcal{J}_n(\hat{\boldsymbol{\eta}}_{\gamma, \delta'})$, $\mathcal{J}_n(\hat{\alpha}_{\gamma, \delta'})$, and the orthogonal projection onto the span $\mathbf{1}_n$ under the information inner product $\mathcal{P}_{\mathbf{1}_n}$, respectively. Regarding the Wald statistic, it is important to note that, since $\mathcal{P}_{\mathbf{1}_n}$ projects onto the span $\mathbf{1}_n$, $(\mathbf{I}_n - \mathcal{P}_{\mathbf{1}_n})\mathbf{1}_n = \mathbf{0}_n$ and $(\mathbf{I}_n - \mathcal{P}_{\mathbf{1}_n})\tilde{\mathbf{X}}_{\gamma, \delta''} = (\mathbf{I}_n - \mathcal{P}_{\mathbf{1}_n})\tilde{\mathbf{X}}_{\gamma, \delta'}\mathbf{D}$. Then

$$\begin{aligned}\mathcal{J}_n(\hat{\boldsymbol{\beta}}_{\gamma, \delta''}) &= \tilde{\mathbf{X}}_{\gamma, \delta''}^\top (\mathbf{I}_n - \mathcal{P}_{\mathbf{1}_n})^\top \mathcal{J}_n(\hat{\boldsymbol{\eta}}_{\gamma, \delta''}) (\mathbf{I}_n - \mathcal{P}_{\mathbf{1}_n}) \tilde{\mathbf{X}}_{\gamma, \delta''} \\ &= \mathbf{D}^\top \tilde{\mathbf{X}}_{\gamma, \delta'}^\top (\mathbf{I}_n - \mathcal{P}_{\mathbf{1}_n})^\top \mathcal{J}_n(\hat{\boldsymbol{\eta}}_{\gamma, \delta'}) (\mathbf{I}_n - \mathcal{P}_{\mathbf{1}_n}) \tilde{\mathbf{X}}_{\gamma, \delta'} \mathbf{D} \\ &= \mathbf{D}^\top \mathcal{J}_n(\hat{\boldsymbol{\beta}}_{\gamma, \delta'}) \mathbf{D}.\end{aligned}$$

Hence:

$$\begin{aligned}Q_{\mathcal{M}_{\gamma, \delta''}} &= \hat{\boldsymbol{\beta}}_{\gamma, \delta''}^\top \mathcal{J}_n(\hat{\boldsymbol{\beta}}_{\gamma, \delta''}) \hat{\boldsymbol{\beta}}_{\gamma, \delta''} \\ &= (\mathbf{D}^{-1} \hat{\boldsymbol{\beta}}_{\gamma, \delta'})^\top (\mathbf{D}^\top \mathcal{J}_n(\hat{\boldsymbol{\beta}}_{\gamma, \delta'}) \mathbf{D}) (\mathbf{D}^{-1} \hat{\boldsymbol{\beta}}_{\gamma, \delta'}) \\ &= \hat{\boldsymbol{\beta}}_{\gamma, \delta'}^\top \mathcal{J}_n(\hat{\boldsymbol{\beta}}_{\gamma, \delta'}) \hat{\boldsymbol{\beta}}_{\gamma, \delta'} = Q_{\mathcal{M}_{\gamma, \delta'}}.\end{aligned}$$

This establishes the equality of the Bayes factors for any two duplicate models of $\mathcal{M}_{\gamma, \delta}$, thereby justifying the use of $\mathcal{M}_{\gamma, \delta}$ as the representative of its full-rank duplicates. In practice, a non-identifiable model supplied to statistical software is automatically replaced by an equivalent full-rank parameterization, and the results above show that the choice of parameterization is irrelevant. Consequently, if model selection identifies an overparameterized model (e.g., the HPM or the MPM), any full-rank parameterization of the corresponding equivalence class may subsequently be used for prediction, interpretation, and inference. Thus, the notation in (26) is slightly abusive, as the model-dependent quantities for the Bayes factor of a non-identifiable model are in fact those of any of its duplicates. Accordingly, the complexity penalty in (26) must also be adjusted, substituting $k_{\gamma, \delta}$ by $k_\gamma + p_{\delta'}$.

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